

Does Cryptocurrency Trading Have Spillover Effects on the Stock Market?*

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Abstract

We investigate whether cryptocurrency trading has spillover effects on the stock market. Using excessive return comovement to identify stocks that are likely traded by cryptocurrency investors, we find evidence of trading-induced price pressure. Stocks that exhibit high excess comovement with cryptocurrencies experience temporary overvaluation and subsequent return reversal of -0.69% per month after netting out known characteristics. The negative return premium increases after periods of high social transmission. Instrumental variable analysis using Tether flows and authorizations supports a causal interpretation of our findings. Our results add to the growing evidence that cryptocurrency traders impact the pricing of other financial assets.

Keywords: Cryptocurrency, Price pressure, Spillover, Stock market, Social transmission

JEL Classification Codes: G11, G12, G14

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1. Introduction

In the United States, cryptocurrency has gone mainstream. Despite the highly volatile nature of the asset class, American households are increasingly adding crypto to their investment portfolios. Adoption is expected to increase further after the U.S. Securities and Exchange Commission recently approved the listing and trading of spot bitcoin exchange traded funds. Many crypto-investors have experienced rapid wealth gains that have created measurable spillover effects on the real economy. For example, Aiello, Baker, Balyuk, Di Maggio, Johnson, and Kotter (2023a) find evidence that wealth effects due to cryptocurrencies increase consumption and local house prices. In this study, we ask whether cryptocurrency trading has spillover effects on the pricing of equities.

Our main hypothesis is that after cryptocurrency investors experience or observe the extreme volatility of cryptocurrency returns, they drive up the prices of other securities with low-probability, high-upside outcomes. It is natural to consider stocks because the majority of cryptocurrency investors also have traditional after-tax brokerage accounts (Aiello et al., 2023b). If their trading is sufficiently large to generate price pressure, these stock could become over-valued and experience predictable return reversals. The empirical challenge is to identify the stocks traded by cryptocurrency investors as they may trade different stocks at different times, with varying intensities. Using a measure of *excess* return comovement, we find that crypto-traders generate non-fundamental demand shocks that produce temporary over-valuation and subsequent return reversals. Additional tests suggest that social transmission bias may partially explain these patterns.

Return comovement between cryptocurrency and individual stocks can arise for two reasons.

First, fundamental cash flow shocks or common risk exposures can simultaneously drive returns in both assets. For example, some companies may have exposure to cryptocurrency due to the nature of their business (e.g., technology), while other companies may adopt a business model that creates direct exposure to cryptocurrency fluctuations (e.g., converting cash reserves to bitcoin). Second, cryptocurrency investors may trade in and out of cryptocurrency and equities for non-fundamental reasons (i.e., noise). Such trading can produce return comovement that is (1) *excessive* in the sense that trading is not driven by cash flow shocks or common risk exposures, and (2) *non-directional*. Our objective is to identify this type of comovement in prices that is not related to comovement with fundamentals (Barberis, Shleifer, and Wurgler, 2005). In the cryptocurrency space, there are economic forces that could create such excessive non-fundamental comovement.

For example, crypto-traders could create excessive positive comovement after receiving a capital endowment, such as government stimulus checks, and concurrently investing the sums in cryptocurrency and equities (Aiello et al., 2023b; Divakaruni and Zimmerman, 2023). Excess negative comovement can also result from cryptocurrency trading strategies that are associated with tax and re-balancing motives. For example, a popular strategy is to sell cryptocurrency to tax-loss harvest (Cong, Landsman, Maydew, and Rabetti, 2023), which investors can then rotate into equities. As a result, the selling pressure in cryptocurrency and the buying pressure in equities can create excess negative comovement. Negative comovement can also arise from differences in the way investors trade cryptocurrencies and equities. Kogan, Makarov, Niessner, and Schoar (2023) find that retail traders use contrarian strategies to trade equities, but momentum-like strategies to trade cryptocurrency. Hence, crypto-traders could create excess negative comovement when they buy contrarian equities

holdings (i.e., recent poor performing stocks) and concurrently sell recent cryptocurrency losers. Moreover, cryptocurrency investors also reallocate funds between crypto and equities with the objective of maintaining relatively stable portfolio weights between asset classes (Aiello et al., 2023a). Such portfolio rebalancing is a form of noise trading that could also induce negative comovement. Overall, trading due to non-fundamental motives can produce both excess positive and negative comovement between cryptocurrency and equities.

To operationalize our analysis, we estimate the return comovement between bitcoin and U.S. individual stocks over the prior three months to create a measure called *BTC sensitivity*, which is the absolute value of the comovement estimate. We chose bitcoin because it is the largest cryptocurrency with the longest history, but our findings are similar using aggregated cryptocurrency market returns. Researchers use signed comovement (i.e., betas) to estimate fundamental or factor exposures, but to measure non-fundamental demand shocks, we require unsigned comovement for the reasons discussed above. Our empirical approach is not new. It is similar in spirit to the literature on market contagion (Bae, Karolyi, and Stulz, 2003) that also uses unsigned comovement to measure spillover between international markets, where investor excitement in one market sparks excitement in other markets, regardless of firm or market fundamentals.

Our spillover hypothesis has two clear and testable implications. The first observable feature is a share price runup due to price pressure. To ensure that *BTC sensitivity* is not commingling fundamental blockchain exposure (beta) with excessive non-fundamental trading, we conservatively drop all stocks that report any use of bitcoin or blockchain technology (Autore, Clarke, and Jiang, 2021).¹ Second, if trader demand bids up prices sufficiently above

¹Our main findings are similar with or without the inclusion of these stocks, which supports the view that

their intrinsic value, then we expect a subsequent predictable return reversal.

Figure 1 illustrates the main result. We create a long-short portfolio that buys high *BTC sensitivity* stocks and shorts low *BTC sensitivity* stocks. Consistent with our prediction, the portfolio experiences a price spike during the initial three-month formation period and a sharp reversal over the next 12 months. Extended return reversal is evidence against possible risk theories or explanations based on common cash flow or discount rate shocks. Using sorts, we find that the long-short portfolio produces a monthly four-factor (i.e., Fama–French three factors plus the momentum factor) alpha of -1.53% (t-stat = -4.61). After confining the sample to only stocks with significant estimates of *BTC sensitivity*, the negative premium of *BTC sensitivity* increases to -2.46% (t-stat = -5.80).

If *BTC sensitivity* captures investor demand for skewness, it might repackage known lottery characteristics and explain our findings. To net out the effects of characteristics related to investor demand for skewness, we develop a methodology to create a binned scatterplot of Fama–MacBeth (1973) (FM) cross-sectional regressions to provide a visual representation of *BTC sensitivity* portfolio returns. Our approach first estimates FM regressions that include five skewness-related characteristics and standard characteristics known to explain the cross-section of stock returns.² Then, we create a binned scatterplot that displays portfolio returns after removing the return premiums associated with known characteristics. Figure 2 shows that high *BTC sensitivity* stocks are overpriced because the negative *BTC sensitivity* premium is driven by the underperformance of stocks in the highest bins of *BTC sensitivity*

fundamental exposure is not the mechanism behind our results.

²The five skewness related characteristics include the price level (Kumar, 2009; Han and Kumar, 2013), idiosyncratic volatility (Ang, Hodrick, Xing, and Zhang, 2006; Hou and Loh, 2016), distress risk (Campbell, Hilscher, and Szilagyi, 2008; Conrad, Kapadia, and Xing, 2014), maximum daily return (Bali, Cakici, and Whitelaw, 2011), and expected skewness (Boyer, Mitton, and Vorkink, 2010).

and not the outperformance of stocks in the lowest bins of *BTC sensitivity*. The difference between the extreme bins generates a monthly portfolio return of -0.69% (t-stat $= -2.60$). After omitting the highest bins, the slope of the scatterplot is flat and is not statistically different from zero.

We also rule out risk-based explanations by estimating factor model regressions using the most prominent risk benchmarks. Across this range of asset pricing tests, we continue to observe a significant and negative *BTC sensitivity* premium.³

To better understand the spillover process, we examine the role of social transmission (Hirshleifer, 2020; Han, Hirshleifer, and Walden, 2022). We suspect that social transmission plays an important role in our setting because social media sentiment drives cryptocurrency returns (Canayaz, Cao, Nguyen, and Wang, 2023). First, we analyze social media activity on Twitter. Among high *BTC sensitivity* stocks, the negative *BTC sensitivity* premium is nearly twice as negative for stocks that have tweets with concurrent mentions of bitcoin compared to those without. For this test, dropping fundamental crypto stocks is useful for distinguishing between fundamental and excessive trading. Second, we extract Google search trends for bitcoin following Liu and Tsyvinski (2021), who use this measure to capture investor attention. We argue that search interest also embodies social transmission from in-person or online discussions because attention is a necessary condition for agents to exchange ideas. When social transmission is high, traders are likely to increase their online search activity and may chase stocks with high *BTC sensitivity*. Consistent with this view, the negative *BTC sensitivity* premium is nearly twice as negative after periods of high search activity compared

³Our study focuses on the spillover effect from cryptocurrencies to equities, but cryptocurrency trading may also draw attention away from lottery-like stocks and produce a substitution effect similar to the patterns documented in Gao and Lin (2015). Although a complete analysis is beyond the scope of this study, we find support for a substitution effect and discuss our findings in Section 6.6

to periods of low search activity. In contrast, following periods of high and low market sentiment (Baker and Wurgler, 2006), we find no significant difference in the negative *BTC sensitivity* premium. The latter result is important because it rules out market sentiment as a possible explanation for the negative *BTC sensitivity* premium and further indicates that *BTC sensitivity* is distinct from known lottery anomalies because the future returns of lottery anomalies are contingent on market sentiment states (Stambaugh, Yu, and Yuan, 2012).

Third, we examine stocks with high retail trading because retail traders are more likely to be affected by social transmission. Consistent with this view, the negative *BTC sensitivity* premium after high search activity is concentrated among stocks with high retail trading. In general, we find that the overpricing of *BTC sensitivity* stocks is significantly higher when social interactions intensify, which supports the view that the spillover of cryptocurrency trading to equities is in part a social phenomenon.

Our asset pricing tests alone do not provide definite conclusions about causality. One concern is that we cannot rule out all common risk or behavioral factors because such factors for cryptocurrencies may have not yet been discovered. Another empirical hurdle is our lack of comprehensive individual account data to examine trading. An ideal experiment to assess causality would be to find episodes of non-fundamental shocks to cryptocurrency prices. Using this ‘noise,’ we can instrument our *BTC sensitivity* measure and test whether it continues to predict future returns. We propose that the controversial Tether stablecoin provides an opportunity to design such a test. Griffin and Shams (2020) show that the authorization and flow of Tether tokens were used to artificially increase the price of bitcoin. Tether activity is an ideal econometric instrument because it perturbs Bitcoin returns through noise—after Tether authorizations, bitcoin appreciates but reverses soon after. We must also

consider whether Tether meets the exclusion restriction for the instrument to be valid. We believe that it does because Tether flows were almost certainly not intended to influence equity prices of the specific set of equities that have high *BTC sensitivity*. Using Tether to identify non-fundamental shocks to bitcoin returns, we find evidence indicative of a causal interpretation.

Additional robustness tests confirm that the main findings are not sensitive to our econometric choices. The negative *BTC sensitivity* premium remains statistically significant when *BTC sensitivity* is estimated using the aggregated cryptocurrency market return or the CoinMarketCap.com subsample. The results are also robust to (i) noise issues, (ii) the inclusion of controls for fundamental exposure to bitcoin, (iii) the exclusion of all technology industry firms, (iv) and the hold-out sample of financial firms. Finally, we find that the *BTC sensitivity* premium is driven by positive estimates (θ^{B+}), which provide suggestive evidence that the tax and rebalancing motives from negative estimates (θ^{B-}) are not the primary cause of overvaluation.

Our study adds to the growing literature on the spillover effects of cryptocurrencies on other asset classes. Aiello et al. (2023b) use transaction-level data from millions of U.S. households' bank accounts and credit card payments to analyze how crypto wealth influences economic activity. They estimate a marginal propensity to consume out of crypto wealth of \$0.08, which is larger than estimates from other asset classes such as housing and equities, but smaller than lottery winnings. They also find that increases in crypto wealth cause significant house price growth. Our findings suggest that cryptocurrency investors affect stock prices through their trading behavior.

Second, our setting provides a unique opportunity to examine the role of social transmission

bias in financial market outcomes (Hirshleifer, 2020; Han, Hirshleifer, and Walden, 2022). The patterns in the data are generally consistent with the predictions of the theory as the relation between *BTC sensitivity* and future returns is dependent on the state of social transmission. More generally, our findings imply that investor behavior can have cross-market effects and add to evidence on gambling attitudes and the stock market (Kumar, 2009; Barber, Lee, Liu, and Odean, 2009; Gao and Lin, 2015).

2. Data and construction of the BTC sensitivity measure

2.1. Data and the BTC sensitivity measure

We collect data from various sources. We obtain trading data for US common share stocks (with a share code of 10 or 11) listed on the NYSE, AMEX, and NASDAQ from the Center for Research in Securities Prices (CRSP). We exclude financial firms (SICs 6000 to 6999) and utility firms (SICs 4900 to 4999) and drop the bottom fifth percentile by market capitalization in each month to avoid bid–ask bounce and issues arising from noise in returns. We perform this filter to ensure that our analysis is comparable with the extant literature. To reduce the chance of capturing fundamental exposure to blockchain, we also exclude all stocks associated with blockchain or bitcoin in news articles (Factiva) or their regulatory 10K/10Q filings following Autore et al. (2021).⁴ In Section 3.1.2 we verify that these filters do not affect our main inferences. We treat delisting returns following Shumway (1997).⁵

We collect daily bitcoin prices from CoinMarketCap.com, which is a widely used website

⁴We obtain the sample of firms from the published article’s Internet Appendix.

⁵Our main findings are similar when using the following stock screens: (i) using all stocks, (ii) excluding stocks with a price lower than \$1 in the previous month, (iii) excluding stocks with a price lower than \$5 in the previous month, (iv) excluding stocks with market cap below the twentieth percentile of NYSE stocks, and (v) using NYSE stocks only. These stock screening methods do not affect our main findings.

that tracks cryptocurrency prices. To ensure that the data are reliable, we cross-check the accuracy of the price data with Coindesk.com. Bitcoin price data are only available after April 28, 2013 on CoinMarketCap.com, so we collect earlier bitcoin price data from Mount Gox. Our sample period covers 147 months, beginning in October 2010 and ending in December 2022. We align daily bitcoin returns to daily stock trading days by calculating the change in price over consecutive NYSE trading days. Our main analysis focuses on bitcoin because it is the most popular cryptocurrency, has the longest history, and represents nearly 50% of the total market capitalization. We also create a top-ten cryptocurrency index and find similar results. The Internet Appendix provides additional descriptions of the bitcoin data.

To construct the *BTC sensitivity* measure, we obtain a parameter estimate $\theta_{i,t}^B$ using the following regression model:

$$\begin{aligned}
 R_{i,t-k} - R_{f,t-k} = & \alpha_{i,t} + \theta_{i,t}^B BTC_{t-k} + \beta_{i,t}^{MKT} MKT_{t-k} \\
 & + \beta_{i,t}^{SMB} SMB_{t-k} + \beta_{i,t}^{HML} HML_{t-k} + \varepsilon_{i,t-k}
 \end{aligned} \tag{1}$$

where $R_{i,t-k}$ is firm i 's daily stock return, and $R_{f,t-k}$ is the daily risk-free rate (30-day U.S. Treasury-bill yield) in the previous k months. BTC_{t-k} is the daily bitcoin return over the risk-free rate. MKT is the market factor (market return over the risk-free rate). SMB is the size factor, and HML is the value factor, as defined in Fama and French (1993).⁶ Because bitcoin has a relatively short history, we set the estimation window k at three months. To ensure reliable estimates, we require at least 50 days of trading data in the three-month estimation window. We include Fama–French (FF) factors such that $\theta_{i,t}^B$ is orthogonal to

⁶We obtain these factors from Kenneth French's website.

priced risks that are known to explain expected stock returns.⁷

As we intend to identify non-fundamental return comovement, we define *BTC sensitivity* as the absolute value of parameter estimate $|\theta_{i,t}^B|$ to capture excess and non-directional comovement. Flows that concurrently trade bitcoin and stocks together in the same direction can create excessive positive estimates (θ^{B+}). Negative estimates (θ^{B-}) can arise when traders switch between bitcoin and stocks, perhaps to chase past performance. Excess comovement may also arise due to a popular strategy that harvests tax loss from cryptocurrencies to offset stock gains because bitcoin is treated as property under U.S. tax law and not subject to wash trading rules (Cong et al., 2023).⁸

We construct firm characteristics for market capitalization (*Market cap*), book-to-market ratio (B/M), asset growth (TAG), operating profitability (OP), past one-year stock return from month $t-12$ to month $t-2$ (MOM), and past month $t-1$ return reversal (STR). We calculate institutional ownership ($IO\%$) from the Thomson Reuters 13F database. We also construct the following measures that proxy for lottery-like features: idiosyncratic volatility ($IVOL$) (Ang et al., 2006), maximum daily return (MAX) (Bali et al., 2011), and expected idiosyncratic skewness (EIS) (Boyer et al., 2010). We proxy for market frictions and illiquidity using daily Turnover and Amihud illiquidity ($ILLIQ$). We create a measure of retail trading activity (*Retail trading*) by identifying retail orders in the TAQ database as those orders that receive price improvement following Boehmer et al. (2021). Specifically, we identify retail trades as trades in the TAQ database with code “D” that clear at a fraction of a penny but

⁷Our results are robust to different estimation horizons (one month, six months, and a principal component analysis factor that combines the one-, three-, and six-month estimates) and adjustments for estimation noise (Asparouhova et al., 2010) and non-synchronous trading (Dimson, 1979).

⁸Wursthorn, M. (2021, October 4). Financial advisors pitch bitcoin to investors to offset portfolio losses. *The Wall Street Journal*.

not around the half penny (0.004, 0.005, 0.006). We winsorize all continuous variables at the 1% and 99% levels to remove the influence of outliers. Appendix A provides variable construction details.

2.2. Characteristics of extreme *BTC sensitivity* stocks

To analyze the characteristics of high-*BTC sensitivity* stocks, we sort stocks each month into 10 groups based on *BTC sensitivity*. Table 1 reports the time-series averages of each stock characteristic across the 10 groups. The final two columns separate the highest *BTC sensitivity* portfolio into θ^{B+} and θ^{B-} , respectively.

Stocks in the highest *BTC sensitivity* decile can be characterized as unprofitable small-cap firms with low institutional ownership, high turnover, and high retail trading activity. While these stocks are relative losers over the past year, they experience relatively high returns in the most recent three months. They have low operating profitability and exhibit lottery-like features (Kumar, 2009) such as stock price level, expected idiosyncratic skewness, maximum daily return, and idiosyncratic volatility. Hence, *BTC sensitivity* may represent a trade-based proxy for expected skewness, which is notoriously difficult to measure (Liu et al., 2022) because expected skewness represents investors' beliefs about tail events that have yet to occur and are usually unobservable to the researcher. From these simple sorts, we observe that stocks in the high *BTC sensitivity* decile experience negative unconditional future returns ($\text{Return } t + 1$) on average. The last two columns confirm that extreme θ^{B+} and θ^{B-} stocks share similar characteristics. The similarity lends credence to our claim that extreme values of θ^B capture correlated demand from non-fundamental trading. We provide further support for this argument in Section 3.

Table 2 presents correlations of *BTC sensitivity* and stock characteristics. The overall patterns are similar to the previous sorts. *BTC sensitivity* exhibits the highest correlations with lottery-type characteristics, including idiosyncratic volatility, expected idiosyncratic volatility, maximum daily return, price, and size.

3. Return patterns of stocks with high BTC sensitivity

3.1. Portfolio sorts on *BTC sensitivity*

We conduct portfolio sorts and examine the returns around the portfolio formation month.

3.1.1. Return patterns around portfolio formation

We sort stocks each month into deciles based on *BTC sensitivity* and calculate the realized returns of each portfolio in the following month. We create a long-short zero-cost portfolio that buys stocks in the top *BTC sensitivity* decile and sells stocks in the bottom *BTC sensitivity* decile. We calculate the returns for the long-short portfolio three months before and twelve months after portfolio formation.

Figure 1 shows the predicted time-series hump-shaped return pattern around the formation month. The long-short portfolio return steadily rises in the pre-formation months and peaks in the formation month. The long-short portfolio returns are significantly negative in the following months after portfolio formation.

3.1.2. Portfolio sorts

Next, we report the average *BTC sensitivity*-sorted portfolio excess returns (raw return minus the risk-free rate) and alphas based on the capital asset pricing model (CAPM),

Fama–French (FF) three-factor, and FF four-factor (including momentum factor) models using equal-weighted portfolios in the immediate month after portfolio formation. We calculate t -statistics based on Newey–West (1987) standard errors with up to 12 lags.

The results in Table 3 compliment Figure 1. We observe that high *BTC sensitivity* stocks suffer significantly low future returns. The first column in the left panel shows that the equal-weighted long-short portfolio generates an average monthly raw return of -1.33% ($t\text{-stat} = -2.67$), or more than -15% per year ($-1.33 \times 12 = -15.96\%$). The next column reports slightly larger abnormal returns of -1.77% ($t\text{-stat} = -4.16$) using the CAPM. The three- and four-factor models in columns (3) and (4) produce similar patterns with average monthly alphas of -1.61% ($t\text{-stat} = -4.71$) and -1.53% ($t\text{-stat} = -4.61$), respectively. A robust return pattern emerges across these specifications. The highest two *BTC sensitivity* deciles consistently exhibit significantly negative risk-adjusted average returns. In contrast, the lowest two *BTC sensitivity* deciles have insignificant risk-adjusted average returns. This suggests that the negative *BTC sensitivity* premium is due to the underperformance of high *BTC sensitivity* stocks and not due to the outperformance of low *BTC sensitivity* stocks. The results are robust to the following alternative specifications: (i) including blockchain-related stocks, (ii) including financial and utility firms, and (iii) using value-weighted portfolios. These results are available in the Internet Appendix (Table IA.2 and Table IA.3, respectively).

3.1.3. Portfolio sorts using only stocks with significant *BTC sensitivity* estimates

The OLS regression model (Equation 1) used to estimate *BTC sensitivity* is likely to incur estimation noise. Therefore, we repeat our portfolio analysis using only stocks that have *BTC sensitivity* estimates at the 10% significance level or better. The return patterns are much

larger using this sample. The right panel of Table 3 reports a monthly four-factor alpha of -2.46% ($t\text{-stat} = -5.80$) for the long-short portfolio, which is 93 basis points larger than the corresponding four-factor alphas using all stocks in the left panel. The larger negative *BTC sensitivity* premium using more precise estimates indicates significant estimation noise in our *BTC sensitivity* measure. The implication is that our baseline tests are downward biased and understate the true magnitude of the expected returns associated with high *BTC sensitivity* stocks.

Overall, the return patterns in Figure 1 and Table 3 are consistent with the view that high *BTC sensitivity* stocks experience price pressure that creates temporary overvaluation and subsequent return reversals.

3.2. Are the BTC sensitivity return patterns due to known characteristics or risks?

If *BTC sensitivity* represents investor demand for skewness, we might be re-documenting the return patterns of lottery-like characteristics in the existing literature. Alternatively, *BTC sensitivity* could capture a known equity risk factor. This section carefully examines this concern using (i) monthly cross-sectional Fama–MacBeth regressions and (ii) factor model tests.

3.2.1. Fama–MacBeth cross-sectional regressions

To investigate the marginal effect of *BTC sensitivity* on expected returns, we first estimate Fama–MacBeth cross-sectional regressions then develop a methodology to create a binned scatterplot (binscatter) from the cross-sectional Fama–MacBeth regressions. To compare coefficient estimates across different specifications, we normalize all variables in each cross-

section to have mean zero and a standard deviation of one. We then report the time-series averages of the coefficient estimates among the sample period. The t-statistics are adjusted following Newey and West (1987) with up to 12 lags.

Table 4 reports the Fama–MacBeth regression results. We observe negative *BTC sensitivity* slopes across all specifications at the 1% significance level. The univariate regression in column (1) produces a coefficient estimate of -0.42 (t-stat = -3.03), which corresponds to the average monthly return spread of greater than 1% per month between the top and bottom decile *BTC sensitivity* portfolios. The economic magnitude of the associated effect is similar to that reported in Table 3 for the univariate sorts.⁹ Columns (2) and (3) include the common characteristics known to explain the cross-section of stock returns: logarithm of market capitalization ($Ln(SIZE)$), logarithm of book-to-market ratio ($Ln(B/M)$), operating profitability (OP), asset growth (TAG), and past return from month $t-12$ to $t-2$ (MOM). The *BTC sensitivity* slope of -0.31 (t-stat = -3.45) in column (3) remains significantly negative with the inclusion of these characteristics. Column (4) includes lottery-type characteristics ($IVOL$, MAX , EIS , $Ln(Price)$, and $Distress$) and measures of trading frictions (turnover and illiquidity). The loadings on the control variables are consistent with the recent literature. The slope estimate on *BTC sensitivity* of -0.14 (t-stat = -3.92) remains statistically significant after including all characteristics.

It is unsurprising to find that the slope on *BTC sensitivity* attenuates with the inclusion of lottery variables because our main hypothesis is that *BTC sensitivity* is an outcome of investors' search for skewness (i.e., low probability, high-upside outcomes), which overlaps

⁹The extreme decile *BTC sensitivity* cutoff point is approximately 1.28 standard deviations from the mean, which corresponds to a long-short *BTC sensitivity* portfolio based on deciles that is approximately 2.56 standard deviations apart. This translates to roughly $-0.42\% \times 2.56 = -1.08\%$.

with lottery-type stocks. Fama and French (2015) show that the attenuation effect arises for two reasons: slope attenuation and variable attenuation. The former effect occurs because of correlations between the new and existing variables. Therefore, we expect to observe slope attenuation because our hypothesis asserts that *BTC sensitivity* and lottery-like returns reflect a preference for skewness. The variable attenuation effect occurs because of turnover in the extreme portfolios. It is important to note that there is still incremental explanatory power in the cross-section of returns. In the Internet Appendix (Table IA.4), we include the complete set of firm characteristics from Table 1 and find that the slope estimate on *BTC sensitivity* remains statistically significant with the inclusion of these controls.

3.2.2. Binned scatterplot graph

To visualize *BTC sensitivity* portfolio returns after netting out the effects of known characteristics, we design a binscatter graph using the following procedure. Each month, we regress the stock return on a set of control variables (column 4 of Table 4) but omit the *BTC sensitivity* measure. From this regression, we obtain the regression residuals (i.e., the return residuals). Next, we regress *BTC sensitivity* on the same set of control variables and obtain the *BTC sensitivity* residuals. Then, we divide the sample of stocks into 25 bins based on their *BTC sensitivity* residuals and calculate the average return residual and the average *BTC sensitivity* residual. Finally, we use the time-series average of the return residual and *BTC sensitivity* residual for each bin to plot the relation between *BTC sensitivity* and return.

Figure 2 reports the binscatter graph. The solid line represents the best fit line using all 25 bins. The slope of the line is significantly negative (-0.14 , $t\text{-stat} = -6.47$) and equivalent to the parameter estimate of *BTC sensitivity* in column 4 of Table 4. We can also use

the resulting binscatter portfolios to estimate a long-short portfolio return. The difference between the extreme bins generates an average monthly portfolio return of -0.69% (t-stat $= -2.60$). We observe that the slope of the line is driven by the highest five *BTC sensitivity* bins as represented by star markers. To quantify this effect, we exclude the top five bins and refit an ordinary least squares (OLS) line using the 20 lower bins. The slope of this line is less than half the magnitude and is statistically insignificant (-0.05 , t-stat $= -1.34$). Consistent with patterns from the univariate sorts, the results suggest that the negative relation between *BTC sensitivity* and stock return is driven by the underperformance of the highest *BTC sensitivity* stocks.

Overall, the Fama–MacBeth regressions provide compelling evidence that the *BTC sensitivity* effect is priced in the cross-section. The negative *BTC sensitivity* premium cannot be fully explained by size, book-to-market, asset growth, profitability, momentum, short-term reversal, lottery-like features, and illiquidity. The evidence further supports our hypothesis that cryptocurrency trading has spillover effects on the stock market.

3.2.3. Factor model analysis

We investigate whether recently proposed factor models can explain the negative *BTC sensitivity* premium because the traditional Fama–French three-factor model is unable to price small stocks with lottery-type features (Fama and French, 1993). We adjust the long-short portfolio returns using the following five factor models: (i) Fama–French five-factor model, (ii) Fama–French five-factor model plus momentum factor, (iii) Frazzini and Pedersen (2014) betting against beta (*BAB*) factor with the four-factor model, (iv) Stambaugh and Yuan (2017) four-factor model with two mispricing-factors (M4), and (v) Bali et al. (2017)

FMAX factor with the four-factor model.¹⁰

Table 5 reports factor loadings and alpha estimates. The first row shows that the long-short portfolio has positive market exposure. The factor loadings from the three- (second row) and four-factor models (third row) show that high *BTC sensitivity* stocks have positive exposure to small-minus-big (*SMB*) and negative exposure to momentum. The five- and six-factor models show that high *BTC sensitivity* stocks have negative exposure to profitability.¹¹ These factor loadings imply that high *BTC sensitivity* stocks share systematic factors (i.e., recent losers and weakly profitable firms) that are known to have lower future returns. Row 6 shows that the long-short portfolio does not load on the *BAB* factor, which implies that leverage constraints are unlikely to explain the negative *BTC sensitivity* premium. Overall, these loadings are consistent with the stock characteristics shown in Table 1.

These models, however, are unable to fully explain the negative *BTC sensitivity* premium. The long-short portfolio continues to produce significant alphas of -1.49% (t-stat = -5.26) and -1.38% (t-stat = -5.12) from the five- and six-factor models, respectively. Because Fama and French (2016) show that the five-factor model captures much of the low returns associated with idiosyncratic volatility and high beta, the significant alphas imply that the *BTC sensitivity* effect is orthogonal to these known effects. The bottom two rows show that the portfolio loads significantly on the M4 and FMAX factors. Using the M4 model, the long-short portfolio loads negatively on both the *MGMT* and *PERF* factors, but the alpha

¹⁰We obtain the Fama–French five factors from Ken French’s website, the M4 factors from Robert Stambaugh’s website, the *BAB* factor from the AQR website, and the *FMAX* factor from Turan Bali’s website. We thank the researchers for making their factors publically available. The Fama–French five-factor model includes factors for profitability (*RMW*) and investment (*CMA*). The *BAB* model has a factor for the low beta anomaly. The M4 model has two mispricing factors that capture managerial decisions (*MGMT*) and firm performance (*PERF*). The *FMAX* factor has a factor for the *MAX* effect.

¹¹Interestingly, once the momentum factor is included, the loading on high-minus-low (*HML*) becomes significantly negative (rows 3 and 5).

estimate of -0.95% (t-stat = -3.23) is similar to the alpha estimate from the four-factor model. This result suggests that while the long-short portfolio has exposure to systematic mispricing factors, the negative *BTC sensitivity* premium is distinct from the common component of return anomalies captured by the M4 factors.

Among the factor models, the M4 and FMAX models are relatively successful in explaining the negative *BTC sensitivity* premium. The alpha estimates from the M4 and FMAX models are -0.95% (t-stat = -3.23) and -0.79% (t-stat = -3.33), respectively. These findings lend credence to the view that *BTC sensitivity* is a trade-based proxy for expected skewness because these factors have a high correlation with lottery-like stocks (Bali et al., 2017; Liu et al., 2018).¹²

3.3. Are the BTC sensitivity return patterns due to omitted risk factors?

We address the possibility that an omitted risk factor explains the negative *BTC sensitivity* premium using two approaches. First, we construct a possible systematic factor associated with the fundamental value of bitcoin. We re-estimate our factor model tests by including the following three bitcoin-related factors: (i) the monthly bitcoin excess return to provide a market-based valuation metric, (ii) the change in bitcoin transactions on the network to capture the usage rate of bitcoin, and (iii) the change in the hash rate of the bitcoin network. The hash rate measures the total processing power of the bitcoin network used to mine bitcoin. Our results are unchanged with the inclusion of these fundamental risk factors, which we report in the Internet Appendix (Table IA.5). Second, we conjecture that

¹²Asness et al. (2020) show that while the FMAX factor is related to lottery demand, the BAB factor is related to lottery demand only because high beta stocks have high correlations with lottery-like stocks. Their decomposition shows that leverage constraints better explain the BAB factor.

cryptocurrency-related risks not captured by our factors are most likely to concentrate among technology firms. Therefore, we repeat our main return analysis by omitting technology firms from the sample and find similar results, which we report in the Internet Appendix (Table IA.6).

4. Spillover effects: Social transmission channel

The spillover from bitcoin to equities may arise from the social transmission of ideas regarding cryptocurrencies. Social transmission bias (Han et al., 2022) occurs when beliefs or preferences are distorted when investors are influenced by discussing trading strategies or by observing convex and volatile returns. In turn, they may adopt these strategies and generate demand for similar securities with high expected skewness.

4.1. Twitter activity

To test for social transmission, we begin by analyzing social media activity on Twitter. We collect all tweets that mention “BTC” or “bitcoin” and identify whether the tweet also mentions any individual stock tickers. Figure IA.2 in the Internet Appendix provides an example of a tweet that simultaneously mentions bitcoin and individual stocks. Our test starts in January 2014 because the number of tweets simultaneously mentioning bitcoin and individual stocks are limited before then. In each month, we split all the stocks into two subsamples: stocks with tweets that simultaneously mention bitcoin and those without. For each subsample, we estimate the negative *BTC sensitivity* premium within these two samples.

Table 6 reports the results. Column (1) reports the raw returns and four-factor alphas of equal-weighted long-short *BTC sensitivity* portfolios based on stocks without tweets that

concurrently mention bitcoin. Column (2) reports the results for stocks with at least one tweet that concurrently mentions bitcoin. The alpha estimate is almost twice that of stocks without a concurrent bitcoin tweet (-2.60% versus -1.31%). In column (3), we use the number of tweets as the portfolio weight (tweet-weighted) and find that the negative *BTC sensitivity* premium increases to -4.01% (t-stat = -4.78) each month. Overall, the results support the view that social transmission may be a channel through which cryptocurrency trading can spillover to equities.

4.2. Google search evidence

Our second measure of social transmission is based on weekly search volume for the word “bitcoin” from Google Trends. Liu and Tsyvinski (2021) use this measure to capture investor attention in cryptocurrencies. We argue that attention is a necessary ingredient for social transmission. We adjust the current week’s search interest by the average search interest during the previous four weeks and create a monthly measure by computing the average of the adjusted weekly search interest in one month. We hypothesize that if the negative *BTC sensitivity* premium arises from social transmission bias, the premium will systematically fluctuate with periods of high and low attention/social transmission. To test this prediction, we perform a conditional portfolio analysis to examine whether the negative *BTC sensitivity* premium is different across the two regimes.

Panel A of Table 7 reports the results. The negative *BTC sensitivity* premium is significantly larger following high Google search periods. Column (1) shows that the raw return of the long-short *BTC sensitivity* (H–L) portfolio following high Google search periods is more than twice as negative compared to periods following low Google search periods

(-1.72% vs. -0.89%). The difference (-0.83% , $t\text{-stat} = -2.17$) is statistically significant. Columns (2) to (4) show that lottery-type anomaly returns (MAX , $IVOL$, and EIS) are also larger following high Google search periods. A possible interpretation of this finding is that lottery anomalies are affected by sentiment in the cryptocurrency market, which we analyze in Section 6.6.

An alternative explanation for these patterns is that high Google search periods capture stock market sentiment because market sentiment is associated with the mispricing of lottery-like stocks (Stambaugh et al., 2012). We examine this possibility using the Baker and Wurgler (2006) sentiment index. Panel B reports the results. The first column shows that the long-short *BTC sensitivity* portfolio has similar returns following high or low periods of market sentiment (-1.17% vs. -1.24%). The difference is economically small (0.07% , $t\text{-stat} = 0.25$) and not statistically different. For the three lottery anomalies, the results are consistent with Stambaugh et al. (2012) as the anomalies are generally larger following high sentiment periods. Together, the results indicate that the negative *BTC sensitivity* premium is conditional on social transmission states but not on market sentiment states. The latter analysis is important to rule out the possibility that market sentiment explains the negative *BTC sensitivity* premium and to provide additional evidence that *BTC sensitivity* is distinct from known lottery anomalies.

4.3. Retail trading activity

Our third test for social transmission focuses on retail trading activity because we expect retail traders to be more likely to be influenced through word-of-mouth communication and social interactions. To identify retail order flow, we collect orders that receive price

improvement following Boehmer et al. (2021). Each month, we split the sample of stocks into two groups based on the median level of retail order flow. Then within each group, we sort stocks into deciles based on *BTC sensitivity* and compute the difference between the top and bottom deciles. We perform this calculation following periods of high and low Google search activity.

Panel C of Table 7 presents the results. The first column reports that the negative *BTC sensitivity* premium is significantly larger among high retail trading stocks following high Google search activity periods (-1.90% , $t\text{-stat} = -6.42$) compared to low Google search activity periods (-0.84% , $t\text{-stat} = -2.04$).¹³ The difference in returns is statistically significant (-1.06% , $t\text{-stat} = -3.69$). Overall, the results in this section support the view that the spillover may arise from the social transmission of ideas regarding cryptocurrencies.

4.4. Return patterns around portfolio formation for different subsamples

The above analysis focuses on the portfolio performance in the first month after the portfolio is established. To provide more support for our social transmission argument, we also investigate portfolio performance around portfolio formation. Specifically, we expect that stocks with social discussion and intense retail trading activity will exhibit stronger return patterns around the formation month. We do not perform this test for our Google search measure because search sentiment lacks cross-sectional heterogeneity across firms.

To visualize the above test, we conduct similar analysis as in Figure 1 by plotting the portfolio performance around the portfolio formation month. The results are presented in

¹³This analysis ends in December 2016 due to the data limitation in our TAQ database. Another useful proxy for retail orders is small trade (trade size below \$5,000) in the TAQ database. However, we cannot use this approach because institutional investors submit much smaller orders in our recent time period (Hu et al., 2018).

Figure 3. We first divide the full sample into subsamples based on whether the stocks are associated with bitcoin-related tweets discussion or whether the stocks have above median retail trading volume. Within each subsample, we create a long-short zero-cost portfolio that buys stocks in the top *BTC sensitivity* decile and sells stocks in the bottom *BTC sensitivity* decile. We then calculate the returns for the long-short portfolio three months before and twelve months after portfolio formation. Consistent with the social transmission argument, Figure 3 shows a stronger hump-shaped return pattern for stocks with tweet discussion (panel A) and stocks with high retail trading activity (panel B). The long-short portfolio return steadily rises in the pre-formation months and peaks in the formation month.

5. Tether flows as an exogenous shock to bitcoin returns

While the evidence supports the view that crypto-trading can spillover to stock prices, some concerns linger. One issue is the possibility of an omitted risk factor because cryptocurrencies may have risks that have yet to be discovered. Another empirical hurdle is our lack of account-level trading data on cryptocurrencies and equities. To assess causality, an ideal experiment would use non-fundamental shocks to bitcoin returns that are plausibly orthogonal to factors that affect stock returns to examine whether the pricing effects we document still exist.

We select the flow and authorization of Tether between different coin exchanges as a possible non-fundamental shock to bitcoin returns. Tether is a cryptocurrency with tokens issued by Tether Limited, whose ultimate owner is Bitfinex. Tether is the largest stablecoin and pegged to the U.S. dollar (USD); in other words, each Tether token claims to be fully

backed by USD reserves.¹⁴ Tether provides a convenient “on-ramp” for many investors to enter and transact in the cryptocurrency market.

Griffin and Shams (2020) show that Tether authorizations and flows between different exchanges predict the future prices of bitcoin and other cryptocurrencies. Bitcoin returns temporarily spike shortly after the authorization and flow of new Tether. Griffin and Shams (2020) argue that the timing of these flows is suspicious and consistent with price manipulation. To use Tether flows as a valid instrument for bitcoin returns, it must satisfy the relevance condition and exclusion restriction. The relevance condition is met if Tether authorizations and flows significantly predict future bitcoin returns. To demonstrate relevance, we replicate the analysis presented in Griffin and Shams (2020) to assess the strength of the instrument. The exclusion restriction is satisfied if Tether authorizations and flows do not affect stock returns other than through *BTC sensitivity*. While the exclusion restriction is fundamentally untestable, we believe it is likely satisfied for the following reason. Tether authorizations and flows are likely used to inflate bitcoin prices, not equity prices. Tether flows were not intended to affect stock prices because cryptocurrencies were traded on exchanges that were separate from equities during our sample period. Moreover, Tether’s founders were likely holding large positions in bitcoin and would have had strong incentives to manipulate or stabilize bitcoin prices rather than equity prices. We obtain Tether flow data from the supplementary materials for the Griffin and Shams (2020) article posted on the Journal of Finance website. The data set contains *hourly* bitcoin returns, bitcoin return volatility, and the Tether authorizations and Tether flows. Following Griffin and Shams (2020), we conduct

¹⁴Tether coin was originally designed to always be worth \$1.00 (i.e., maintaining \$1.00 in reserves for each Tether issued). But an investigation conducted by the New York attorney general found that “Tether’s claims that its virtual currency was fully backed by U.S. dollars at all times was a lie.” <https://ag.ny.gov/press-release/2021/attorney-general-james-ends-virtual-currency-trading-platform-bitfinexs-illegal>.

our first-stage regression at the hourly frequency as follows:

$$\begin{aligned}
\frac{1}{3} \sum_0^2 BTC_{h+i} = & \beta_1 Flow_{h-1} + \beta_2 Auth_{Post(h-1)} + \beta_3 Flow_{h-1} \times Auth_{Post(h-1)} \\
& + \beta_4 BTC_{h-1} + \beta_5 VoB_{h-1} + \beta_6 BTC_{h-1} \times VoB_{h-1} \\
& + \beta_7 MKT_{t-1} + \beta_8 SMB_{t-1} + \beta_9 HML_{t-1} + \varepsilon_h
\end{aligned} \tag{2}$$

where BTC_h is the hourly bitcoin return of an equal-weighted price index that aggregates bitcoin prices on the following exchanges: Bitfinex, Poloniex, Bittrex, Binance, HitBTC, Huobi, and OKEx. The dependent variable is the average hourly return in the next three hours. $Flow_h$ is the average net hourly flow of Tether from Bitfinex to Poloniex and Bittrex and of bitcoin from Poloniex and Bittrex to Bitfinex. $Auth_{post}$ is an indicator variable that equals one if a Tether authorization occurred in the previous 72 hours, and zero otherwise. VoB_h is volatility calculated using hourly bitcoin returns over the previous 24 hours. MKT_{t-1} , SMB_{t-1} , and HML_{t-1} are the daily-level Fama–French three factors, lagged by 1 day. We use the Fama–French three factors in the previous day to prevent a look-ahead bias but verify that our findings are similar using the same-day Fama–French factors.

Table 8 reports the results from the first-stage regression (Equation 2). The results support the view that the instruments satisfy the relevance condition. Consistent with GS, the loading on the interaction of Tether authorization and higher Tether flow (coef.=4.25) is significantly positive with a t-statistic of 2.53, and the control variables also show the correct signs. The Cragg–Donald Wald F-statistic is 9.28 and above the 10% cutoff level for weak instruments for the standard two-stage least squares (2SLS) regression (Stock and Yogo, 2005). We should also draw attention to two special circumstances of our setting.

First, in 2SLS, the economic outcome of interest is the estimated local average treatment effect. In contrast, after we obtain the instrumented parameter estimate on the predicted BTC return in the second stage, we compute the absolute value of that estimate to produce the *BTC sensitivity* measure. Then, our analysis has a third stage where we rank order the instrumented coefficients to examine the relation between *BTC sensitivity* and future stock returns. Second, the concern with weak instruments is that they produce too little variation in predicted outcomes, which causes inconsistent estimates. But in our case, the lack of variation due to weak instruments would create noise in the rank orderings, which would weaken any association between *BTC sensitivity* and returns and potentially biases against finding results.

To obtain the instrumented *BTC sensitivity* measure, we estimate Equation 3.

$$\begin{aligned}
R_{i,t-k} - r_{i,t-k} = & \alpha_{i,t} + \theta_{i,t}^{IB} \widehat{BTC}_{t-k} + \beta_{i,t}^{MKT} MKT_{t-k} + \beta_{i,t}^{SMB} SMB_{t-k} \\
& + \beta_{i,t}^{HML} HML_{t-k} + \beta_{i,t,5} BTC_{t-k-1} + \beta_{i,t,6} VoB_{t-k-1} \\
& + \beta_{i,t,7} BTC_{t-k-1} \times VoB_{t-k-1} + \varepsilon_{i,t-k}
\end{aligned} \tag{3}$$

where $R(i, t - k)$ is firm i 's stock daily return, and $r(f, t - k)$ is the daily risk-free rate (30-day U.S. Treasury-bill yield) in the previous k months. $\widehat{BTC}_{t-k}$ is the instrumented daily bitcoin return over the risk-free rate, which we obtain by compounding the hourly predicted return to the daily level. We include the following control variables from Equation 2, but aggregate them into daily level. BTC_{t-k-1} is the lagged realized daily bitcoin return over the risk-free rate. VoB_{t-k-1} is the lagged average hourly bitcoin return volatility in day $(t-k-1)$. After obtaining the coefficient estimate $\theta_{i,t}^{IB}$, we compute the absolute value to produce the

instrumented *BTC sensitivity* measure.

Figure 4 shows the portfolio performance around the portfolio-formation date. We report the subsample results based on retail trading activities and tweets. For each testing sample, we create a long-short portfolio that buys stocks with high instrumented *BTC sensitivity* and shorts stocks with low instrumented *BTC sensitivity*. Consistent with our baseline result in Figure 1, the portfolio experiences a price runup during the initial three-month period and then a subsequent reversal. The return pattern supports a causal interpretation between the stock's *BTC sensitivity* and further returns.

The caveat to our analysis is that our sample period is short. The GS sample spans from March 1, 2017, to March 31, 2018, which limits our analysis to 11 months (June 2017 to April 2018). Hence, the statistical analysis will lack power. Therefore, we interpret the patterns in Figure 4 as evidence consistent with a causal interpretation.

6. Additional tests

This section reports additional tests to address concerns related to our sample, data selection, estimation issues, and noise-related issues.

6.1. Using the *CoinMarketCap.com* sample

Our main analysis uses bitcoin data from both *CoinMarketCap.com* and Mt. Gox exchange. Using data from *CoinMarketCap.com* (August 2013 to December 2022), we repeat the portfolio analysis in Table 3 and find similar patterns in the long-short portfolio returns. The results are available in the Internet Appendix (Table IA.7).

6.2. Using the aggregated cryptocurrency market return

The cryptocurrency market has expanded beyond bitcoin in recent years. Although bitcoin remains the predominant security (Hu et al., 2019), we verify that our results are robust by repeating the main analysis using a *BTC sensitivity* measure estimated with the aggregated cryptocurrency market return. Specifically, we construct a value-weighted cryptocurrency market return using the market capitalization weighted return of the top 10 cryptocurrencies from CoinMarketCap.com based on their market capitalizations as of March 31, 2018. Our inferences are unchanged using this approach. The results are available in the Internet Appendix (Table IA.8).

6.3. Addressing noise in estimation and prices

We briefly discuss the issues that arise from estimation noise and noise in prices. Estimating the sensitivity of stock returns to bitcoin returns using OLS incurs estimation noise. One concern is that the sensitivities estimated over a three-month horizon could be too short or too long with respect to time-varying sensitivities. Second, non-synchronous trading can bias OLS estimates (Dimson, 1979). Third, because CRSP prices and returns contain non-negligible noise, our return estimates could be biased. We perform the following tests to address these issues. First, we show that our main findings are similar when estimating sensitivities over a shorter one-month horizon or a longer six-month horizon. As an alternative approach, we extract a principal component analysis factor from *BTC sensitivity* sensitivities estimated from the one-, three-, and six-month horizons and use the first component as our measure of *BTC sensitivity*. Our conclusions are unchanged using these measures. The

results are available in the Internet Appendix (Table IA.9). Second, we estimate sensitivities using the Dimson (1979) approach by including a lead and a lag of the bitcoin return and summing together the sensitivity estimates. The results are similar using this approach and are available in the Internet Appendix (Table IA.10). Third, we address concerns regarding noise in returns by weighting stocks by their past month returns, following the guidance of Asparouhova et al. (2010). The results are available in the Internet Appendix (Table IA.11). Overall, the findings of these tests help to rule out the possibility that noise is behind our findings.

6.4. Out-of-sample analysis: Financial firms

Our main analysis excludes financial stocks to remain comparable with the existing literature. As an out-of-sample test, we analyze financial stocks and continue to find a negative *BTC sensitivity* premium among financial firms (see the Internet Appendix, Table IA.12).

6.5. Positive BTC sensitivity versus negative BTC sensitivity

The *BTC sensitivity* measure contains both θ^{B+} and θ^{B-} stocks. We examine the relative importance of each type by separately analyzing positive and negative *BTC sensitivity* subsamples. Cross-sectional FM regression tests suggest that negative *BTC sensitivity* premium is driven by positive *BTC sensitivity* stocks. Although we do not have a definitive explanation for this result, one interpretation is that θ^{B-} generated from tax and re-allocation motives produces weaker pricing effects than flow trading, perhaps due to the costs of moving capital from crypto exchanges to traditional brokerages. The results are available in the Internet Appendix Table IA.13.

6.6. *Do cryptocurrencies draw interest away from lottery stocks?*

Our study examines a spillover effect of cryptocurrencies to equities, but there could be a concurrent substitution effect where cryptocurrencies draw interest away from lottery stocks. To test this possibility, we estimate comovement between lottery stocks and bitcoin. When the comovement is negative, we argue that the substitution effect dominates, which draws interest away from lottery stocks and reduces the negative lottery premium. We find evidence consistent with this prediction (see the Internet Appendix, Table IA.14).

7. Conclusion

Cryptocurrencies are gaining widespread adoption among households in the United States. This paper shows that crypto-trading has spillover effects on equity prices. We identify stocks that are likely to be popular among cryptocurrency investors and demonstrate that crypto-trading induced price pressure creates temporarily overvaluation and predictable return reversals. Our tests using Tether flows as an instrument support a causal interpretation of our findings. In future research, it would be interesting to examine whether cryptocurrencies affect money-related products such as lottery participation, casino activities, and other financial assets such as venture capital and debt markets.

Our setting also provides a unique opportunity to examine the role of social transmission on asset prices (Hirshleifer, 2020; Han et al., 2022). Cryptocurrencies are an attractive venue to examine the role of social transmission because the volatility and skewness of cryptocurrency returns are likely to grab the attention of investors and influence their behavior. Consistent with social transmission effects, the negative *BTC sensitivity* premium is stronger (i) among

stocks that have concurrent bitcoin Twitter mentions and (ii) following periods of high Google search activity, particularly for stocks with high retail trading.

Our findings introduce a new stylized fact on cryptocurrencies. While cryptocurrencies seem to have little correlation with other types of assets at the aggregate level, we show that strong correlations exist in specific segments of the stock market. We document one type of spillover effect, where crypto-investors can push prices away from fundamental value. However, as cryptocurrencies become more integrated with the rest of the traditional financial system, there is potential for other types of spillover. For example, we briefly examine whether the popularity of cryptocurrencies could divert attention and interest away from lottery stocks. We leave the study of these possibilities for future research.

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Appendix A: Variable definitions

- *BTC sensitivity* is estimated using Equation (1) using a three-month period. Bitcoin returns are calculated using the stock market trading day buy-and-hold returns to match the stock returns in the CRSP. The absolute value of the coefficient θ^B is defined as the individual stocks' BTC sensitivity.
- *BTC sensitivity (%Sig.)* is defined as the percentage of stocks with significant BTC sensitivity estimate at the 10% level.
- *No. of stocks* is the average number of stocks in each portfolio.
- *Return $t+1$* is the individual stock's return (in percentage) in month $t+1$.
- *Mkt cap* is the individual stocks' market capitalization (in \$billions) in month t .
- *IO%* is the shares held by institutional investors in quarter $t-2$.
- *MOM* is the cumulative stock return in previous one year except the last one month, i.e., from month $t-2$ to month $t-12$.
- *STR* is the short-term reversal, defined as the previous month's stock return.
- *B/M* is the book value of equity to market value of equity ratio in fiscal year $t-1$.
- *OP* is the annually operating profitability in fiscal year $t-1$, adjusted with R&D expense following Ball et al. (2015).
- *TAG* is the growth rate of the total asset.
- *TO (%)* is the average monthly turnover ratio of the individual stock during the previous 3 months.
- *Retail trading (%)* is defined as the total retail trading volume scaled by the total trading volume in month t . We adjust the CRSP reported trading volume for NASDAQ stocks following Anderson and Dyl (2005).
- *ILLIQ* is the Amihud (2002) illiquidity measure in month t .
- *Price* is the individual stock's price in month t .
- *IVOL* is the idiosyncratic volatility, defined as the standard deviation of the residuals from the regression of daily stock returns to Fama-French three factors in month t .
- *MAX* is the maximum daily stock return in month t .
- *EIS* is the future expected idiosyncratic skewness estimated in month t , as in Boyer et al. (2010).
- *Distress* is the estimated distress risk following Campbell et al. (2008).
- *Flow($h-1$)* is the lagged average net *hourly flow* of Tether from Bitfinex to Poloniex and Bittrex and of bitcoin from Poloniex and Bittrex to Bitfinex.
- *AuthPost($h-1$)* is an indicator variable that equals one if a Tether authorization occurred in the previous 72 *hours* of the time h 's Bitcoin price, and zero otherwise.
- *BTC($h-1$)* is the lagged realized *hourly* bitcoin return.
- *VOB($h-1$)* is Bitcoin price volatility calculated using *hourly* Bitcoin returns over the previous 24 hours.

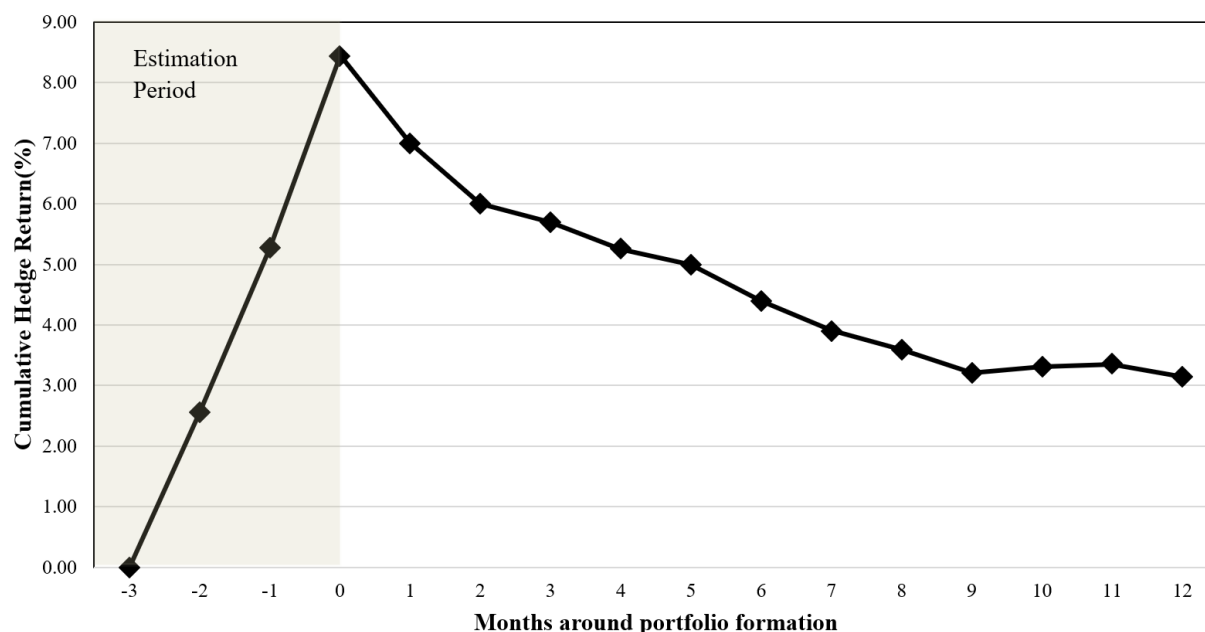


Figure 1: Cumulative long-short BTC sensitivity portfolio returns around the formation month

This figure plots the cumulative long-short *BTC sensitivity* portfolio return in the three months before and twelve months after portfolio formation. The long-short portfolio is a zero-cost portfolio that buys stocks in the highest *BTC sensitivity* decile stocks and shorts stocks in the lowest *BTC sensitivity* decile. In the formation month ($t = 0$), all stocks are ranked and assigned to one of the ten decile portfolios based on their *BTC sensitivity* measure. The sample period is from October 2010 to December 2022.

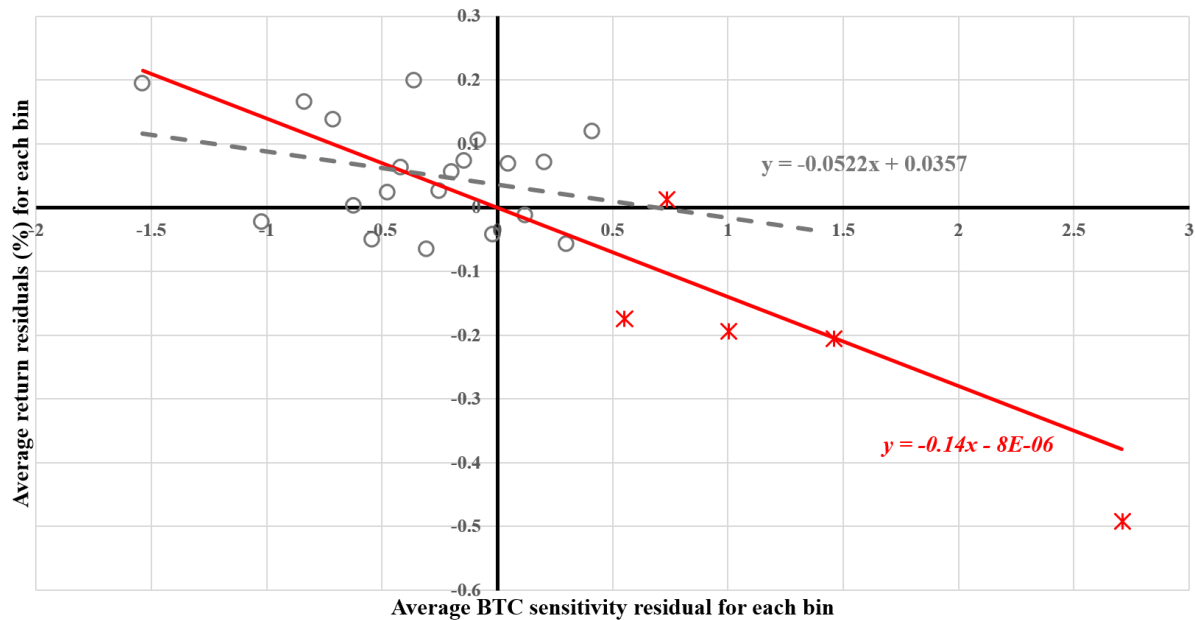
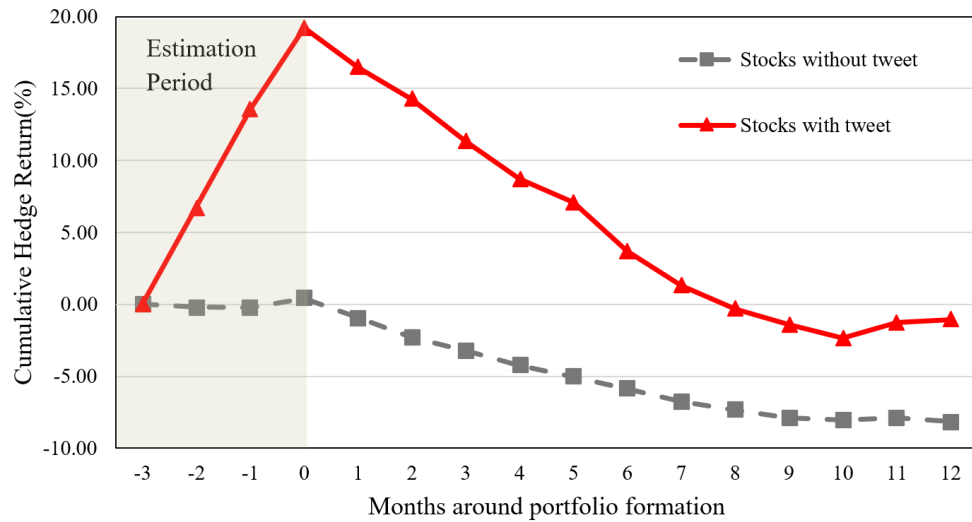
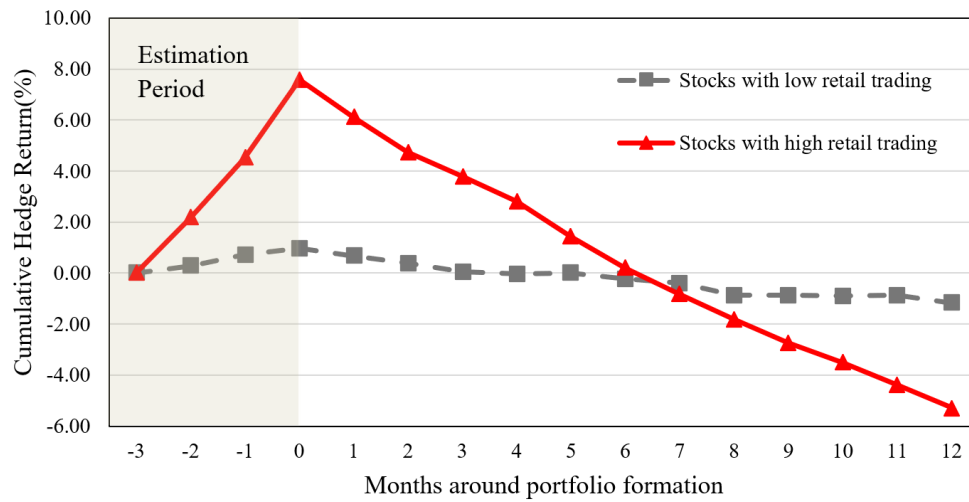


Figure 2: Binned scatterplots of portfolio returns and stock BTC sensitivity

This figure shows a binscatter plot of the relation between portfolio returns and *BTC sensitivity*. Each month, we divide the entire sample into 25 bins based on their *BTC sensitivity* residuals and estimate the average return residual for each bin. Both return residuals and *BTC sensitivity* residuals are obtained by regressing the variable on all control variables. We compute the time-series average of the *BTC sensitivity* residuals and return residuals for the 25 bins and plot the graph. The grey circle represents the first 20 bins, and the red stars represents the last 5 bins. The grey dash line is the OLS fitted line between *BTC sensitivity* residual and return residual using the first 20 bins, and the red line is the fitted line using all 25 bins. The corresponding regression coefficients are also reported along the two fitted lines.



Panel A. Twitter



Panel B. Retail Trading

Figure 3: Cumulative long-short *BTC sensitivity* portfolio return: Social transmission

This figure plots the cumulative long-short *BTC sensitivity* portfolio return in the three months before and twelve months after portfolio formation for different subsamples. Panel A presents the results for sub-samples based on whether the stock had a tweet that simultaneously mentions Bitcoin or those that did not. Panel B presents results for stocks with high or low retail trading activities.

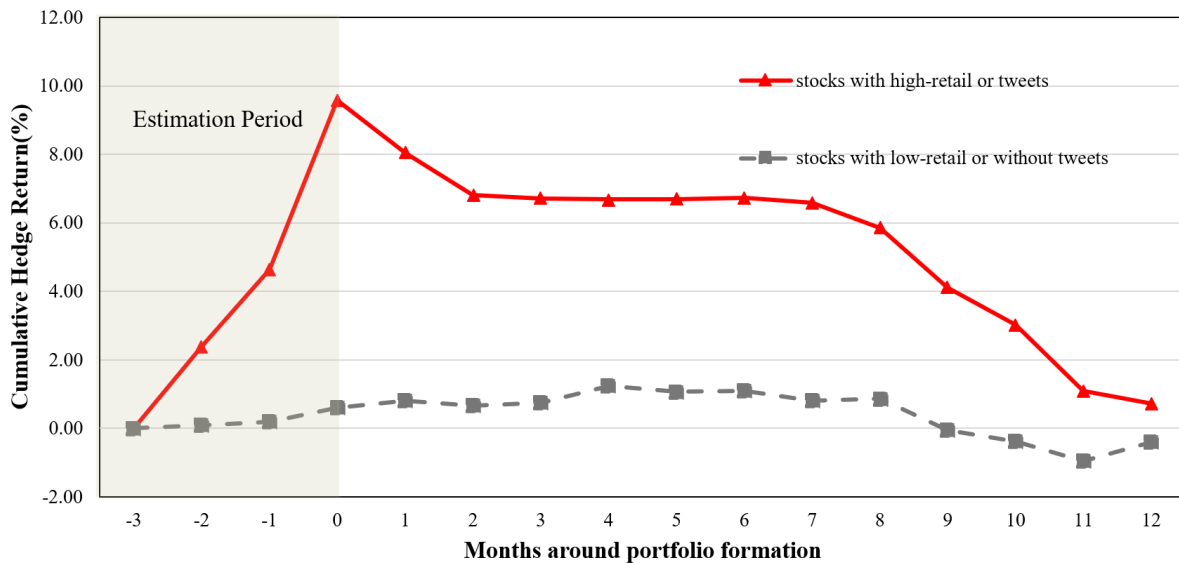


Figure 4: Cumulative long-short portfolio return using the instrumented *BTC sensitivity*

This figure shows the cumulative long-short portfolio returns in the three months before and twelve months after portfolio formation. Two sets of long-short portfolio returns are reported, including subsample results based on stocks with high-retail trading activities or with tweets, and subsample result based on stocks with low-retail trading activities or without tweets. The long-short portfolio is a zero-cost portfolio that longs the highest instrumented *BTC sensitivity* decile stocks and shorts the lowest instrumented *BTC Sensitivity* decile stocks. In the formation month (month t), all stocks are ranked and assigned to one of the ten decile portfolios based on their instrumented *BTC Sensitivity* measure. The sample period is June 2017 to April 2018 due to Tether data availability.

Table 1: Average stock characteristics for portfolios of stocks sorted by θ^B

This table shows the time-series average of the mean values of stock characteristics on ten portfolios sorted by BTC sensitivity. We also report the stock characteristics for the extreme positive BTC sensitivity stocks (top 5%) and extreme negative BTC sensitivity stocks (bottom 5%) in the last two columns. The sample period is from October 2010 to December 2022.

	Low	2	3	4	5	6	7	8	9	High	H-L	θ^B+	θ^B-
<i>BTC sensitivity</i>	0.00	0.01	0.02	0.03	0.04	0.05	0.07	0.09	0.13	0.27	0.27	0.27	0.27
<i>BTC sensitivity (% sig)</i>	0.07	0.71	0.38	0.30	1.17	4.20	9.16	15.72	25.79	48.79	48.72	0.49	0.47
θ^B	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.03	0.03	0.27	-0.27
No. of stocks	252	252	252	252	252	252	252	252	252	252	0.09	135	117
Return t+1 (%)	1.06	0.93	0.93	0.97	0.92	0.99	0.77	0.69	0.37	-0.27	-1.33	-0.57	0.09
Mkt cap (\$B)	2.89	2.95	2.98	2.82	2.58	2.27	1.94	1.53	1.09	0.64	-2.25	0.64	0.64
IO%	65.74	65.62	65.3	64.87	63.68	62.09	60.29	56.66	51.04	40.27	-25.47	39.61	41.1
MOM (%)	14.54	14.49	13.84	13.73	13.37	13.21	12.58	11.43	9.91	8.02	-6.51	9.07	8.33
STR (%)	1.05	1.13	0.91	0.87	0.93	0.87	0.93	0.84	0.76	1.70	0.66	1.83	1.92
B/M	0.55	0.55	0.55	0.56	0.57	0.58	0.59	0.60	0.63	0.66	0.11	0.65	0.67
OP	0.12	0.12	0.12	0.11	0.11	0.10	0.09	0.07	0.04	-0.03	-0.15	-0.03	-0.02
TAG	0.27	0.25	0.22	0.24	0.23	0.25	0.29	0.32	0.37	0.46	0.19	0.45	0.48
TO (%)	13.74	14.07	14.29	14.41	14.82	15.40	16.21	17.14	18.91	28.15	14.41	28.29	27.83
Retail trading (%)	7.56	7.63	7.85	8.01	8.43	9.01	9.54	10.77	12.50	16.10	8.55	16.34	15.84
ILLIQ (10-5)	0.86	0.70	0.56	0.52	0.58	0.59	0.62	0.74	0.91	1.21	0.35	1.18	1.30
Price (\$)	39.94	40.95	41.09	39.40	36.76	33.32	29.54	24.19	18.29	12.07	-27.87	11.99	12.38
IVOL (%)	1.67	1.78	1.88	1.95	2.06	2.21	2.39	2.67	3.12	4.32	2.65	4.32	4.34
MAX (%)	4.75	5.12	5.34	5.54	5.81	6.20	6.66	7.36	8.59	12.11	7.35	12.16	12.13
EIS	0.65	0.66	0.69	0.73	0.79	0.88	0.99	1.16	1.41	1.83	1.18	1.85	1.79
Distress	-8.02	-8.00	-7.94	-7.93	-7.85	-7.79	-7.70	-7.52	-7.25	-6.64	1.38	-6.67	-6.62

Table 2: Time-series average of cross-sectional correlations

This table presents the time-series averages of cross-sectional correlations between *BTC sensitivity* and stock characteristics in Table 1. Variable definitions are available in Table 1. $\ln(\text{Price})$ is the logarithm of stock price in month t . $\ln(\text{Size})$ is the natural logarithm of the market value in month t . $\ln(B/M)$ is the natural logarithm of the book-to-market ratio in month t . Spearman correlations are shown above the diagonal; Pearson correlations are shown below the diagonal. The sample period is October 2010 to December 2022.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
BTC sensitivity	(1)		0.43	0.31	0.37	-0.31	0.07	0.23	-0.28	0.02	-0.04	-0.13	-0.20	-0.02	0.18	-0.20	0.30
IVOL	(2)	0.47		0.58	0.83	-0.56	0.08	0.44	-0.50	0.04	-0.04	-0.24	-0.36	-0.03	0.31	-0.36	0.56
EIS	(3)	0.35	0.52		0.43	-0.74	-0.21	0.72	-0.77	0.21	-0.14	-0.40	-0.39	-0.12	0.38	-0.50	0.65
MAX	(4)	0.39	0.86	0.37		-0.43	0.11	0.34	-0.37	0.03	0.21	-0.21	-0.30	-0.02	0.25	-0.27	0.44
$\ln(\text{Price})$	(5)	-0.34	-0.51	-0.74	-0.37		0.20	-0.74	0.81	-0.25	0.15	0.38	0.45	0.17	-0.45	0.56	-0.70
TO	(6)	0.19	0.24	-0.01	0.22	0.04		-0.62	0.42	-0.16	0.01	0.02	0.06	0.06	0.08	0.39	-0.28
ILLIQ	(7)	0.04	0.08	0.22	0.04	-0.15	-0.15		-0.94	0.28	-0.07	-0.25	-0.39	-0.11	0.26	-0.63	0.68
$\ln(\text{Size})$	(8)	-0.29	-0.44	-0.74	-0.31	0.79	0.18	-0.26		-0.30	0.12	0.29	0.41	0.13	-0.31	0.62	-0.72
$\ln(B/M)$	(9)	0.01	0.01	0.15	0.01	-0.21	-0.06	0.11	-0.28		0.00	-0.05	-0.22	-0.22	0.25	-0.11	0.10
STR	(10)	0.01	0.09	-0.11	0.35	0.11	0.02	-0.01	0.08	0.01		0.04	0.05	-0.01	-0.09	0.05	-0.06
MOM	(11)	-0.08	-0.14	-0.33	-0.13	0.30	0.03	-0.05	0.22	-0.05	0.02		0.14	0.00	-0.43	0.12	-0.20
OP	(12)	-0.25	-0.35	-0.44	-0.28	0.46	-0.07	-0.06	0.38	-0.03	0.02	0.09		0.07	-0.46	0.33	-0.33
TAG	(13)	0.06	0.08	0.04	0.06	-0.01	0.03	-0.01	-0.02	-0.12	-0.02	-0.02	-0.08		-0.08	0.05	-0.08
Distress	(14)	0.16	0.23	0.34	0.17	-0.38	0.13	0.12	-0.31	0.27	-0.02	-0.24	-0.33	-0.01		-0.10	0.15
IO%	(15)	-0.24	-0.36	-0.56	-0.26	0.57	0.16	-0.23	0.60	-0.09	0.02	0.07	0.36	-0.06	-0.18		-0.66
Retail	(16)	0.32	0.49	0.70	0.37	-0.72	-0.09	0.30	-0.71	0.09	-0.02	-0.15	-0.39	0.03	0.18	-0.68	

Table 3: Portfolio sorts: Excess returns and alphas on portfolios of stocks sorted by BTC sensitivity

The table reports the average returns or alphas in percentage of monthly rebalanced decile portfolios sorted by the *BTC sensitivity* measure. The left four columns present results using all stocks with a *BTC sensitivity* estimate. The right four columns report the result using only stocks with a significant regression coefficient at the 10% level. Excess return is the average returns minus the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include the CAPM, the Fama-French three-factor model, and a four-factor model that includes Fama-French three factors and momentum factor. Long-short return (H–L) or alpha is the return or alpha of a zero-cost portfolio that buys the top *BTC sensitivity* decile portfolio and shorts the bottom *BTC sensitivity* decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	BTC sensitivity				Significant BTC sensitivity estimate			
	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
1 Low	1.06 (2.76)	−0.02 (−0.10)	0.16 (1.26)	0.22 (1.51)	1.20 (4.03)	0.35 (1.39)	0.45 (2.59)	0.44 (2.36)
2	0.93 (2.12)	−0.19 (−1.14)	−0.03 (−0.25)	0.00 (0.01)	1.00 (2.58)	−0.16 (−0.49)	−0.01 (−0.07)	−0.04 (−0.18)
3	0.93 (2.03)	−0.31 (−1.76)	−0.12 (−1.58)	−0.06 (−0.86)	1.17 (2.67)	0.01 (0.04)	0.18 (0.96)	0.28 (1.28)
4	0.97 (1.97)	−0.30 (−1.48)	−0.10 (−1.21)	−0.07 (−0.80)	1.28 (2.58)	−0.01 (−0.02)	0.22 (0.96)	0.20 (0.83)
5	0.92 (1.68)	−0.40 (−1.99)	−0.19 (−2.02)	−0.14 (−1.51)	0.88 (1.84)	−0.32 (−0.99)	−0.08 (−0.38)	0.00 (0.00)
6	0.99 (1.87)	−0.34 (−1.97)	−0.11 (−1.40)	−0.05 (−0.63)	0.43 (0.72)	−0.90 (−3.40)	−0.63 (−3.09)	−0.51 (−2.54)
7	0.77 (1.38)	−0.56 (−2.46)	−0.33 (−2.57)	−0.26 (−2.07)	0.26 (0.34)	−1.11 (−2.43)	−0.83 (−2.36)	−0.79 (−1.21)
8	0.69 (1.11)	−0.67 (−2.50)	−0.41 (−2.47)	−0.31 (−1.87)	0.20 (0.21)	−1.46 (−2.85)	−1.12 (−3.26)	−1.06 (−2.99)
9	0.37 (0.54)	−1.04 (−3.34)	−0.71 (−3.94)	−0.59 (−3.83)	−0.34 (−0.36)	−1.93 (−3.29)	−1.59 (−2.89)	−1.57 (−2.76)
10 High	−0.27 (−0.33)	−1.79 (−4.31)	−1.44 (−4.74)	−1.31 (−4.64)	−0.92 (−1.00)	−2.60 (−5.44)	−1.21 (−5.26)	−2.02 (−5.44)
H – L (t-stat)	−1.33*** (−2.67)	−1.77*** (−4.16)	−1.61*** (−4.71)	−1.53*** (−4.61)	−2.12*** (−2.84)	−2.94*** (−5.59)	−2.65*** (−5.45)	−2.46*** (−5.80)

Table 4: Firm-level Fama-MacBeth cross-sectional return regressions

This table reports the time-series monthly averages of coefficient estimates from Fama-MacBeth cross-sectional regressions. We standardize all stock characteristics ($N \sim (0,1)$) for ease of comparison. The dependent variable is a firm's monthly return (in percentage). The control variables include the logarithm firm of market capitalization ($Ln(SIZE)$), the logarithm of book-to-market equity ($Ln(B/M)$), annually operating profitability (OP), total asset growth (TAG), momentum (MOM), idiosyncratic volatility ($IVOL$), maximum daily return (MAX), expected idiosyncratic skewness (EIS), Amihud illiquidity ($ILLIQ$), average daily turnover ratio ($Turnover$), the logarithm of stock price ($Ln(Price)$), and distress risk ($Distress$). The t-statistics (in parentheses) are computed using Newey and West (1987) standard errors with up to twelve lags. ***, **, and, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is October 2010 to December 2022.

	(1)	(2)	(3)	(4)
BTC sensitivity	-0.42*** (-3.03)	-0.36*** (-3.60)	-0.31*** (-3.45)	-0.14*** (-3.92)
Ln(Size)		0.10 (0.70)	-0.02 (-0.18)	-0.07 (-0.80)
Ln(B/M)		0.11 (0.75)	0.07 (0.52)	-0.03 (-0.21)
MOM		0.32*** (3.27)	0.30*** (3.20)	0.28*** (2.95)
OP			0.35*** (3.10)	0.33*** (4.27)
TAG			-0.16** (-1.99)	-0.14* (-1.76)
IVOL				-0.47*** (-4.43)
MAX				0.11 (1.08)
EIS				-0.47*** (-2.97)
Ln(Price)				-0.27 (-1.16)
Distress				0.24*** (3.33)
ILLIQ				-0.03 (-0.96)
Turnover				-0.21** (-2.12)

Table 5: Factor model tests

This table reports reports risk-adjusted returns (alphas) in percentage and the respective factor loadings of the long-short decile portfolio based on BTC sensitivity using eight factor models: (1) the CAPM, (2) the Fama-French three-factor model, (3) the FF3 + momentum-factor model, (4) the Fama-French five-factor model, (5) the FF5 + momentum six-factor model, (6) the Frazzini and Pedersen (2014) betting against beta factor (BAB) + four-factor model, (7) the Stambaugh and Yuan (2017) four-factor mispricing (M4) model, and (8) the Bali et al. (2017) Max factor (FMAX) + four-factor model. The t-statistics (in parentheses) are computed using Newey and West (1987) standard errors with up to twelve lags. ***, **, and, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022. For the M4 mispricing pricing model, the sample period ends in December 2016 due to data availability.

Model	Alpha	MKT	SMB	HML	UMD	RMW	CMA	BAB	MGMT	PERF	FMAX
CAPM model	−1.77*** (−4.16)	0.43*** (6.06)									
3-Factor model	−1.61*** (−4.71)	0.30*** (3.58)	0.81*** (3.33)	−0.21*** (−2.99)							
4-Factor model	−1.53*** (−4.61)	0.27*** (2.85)	0.79*** (3.06)	−0.26*** (−2.81)	−0.14 (−0.79)						
5-Factor model	−1.49*** (−5.26)	0.37*** (3.17)	0.49** (2.43)	−0.26*** (−2.70)		−0.83*** (−5.05)	0.24 (0.89)				
6-Factor model	−1.38*** (−5.12)	0.32** (2.42)	0.45** (2.23)	−0.37*** (−4.79)	−0.22 (−1.55)	−0.87*** (−5.24)	0.32 (1.33)				
BAB model	−1.62*** (−4.76)	0.25*** (2.66)	0.79*** (3.24)	−0.27*** (−3.05)	−0.20 (−1.31)			0.18 (0.70)			
M4 model	−0.95*** (−3.29)	−0.01 (−0.04)	−0.18 (−0.94)						−0.55*** (−4.24)	−0.52*** (−5.84)	
FMAX model	−0.79*** (−3.33)	−0.05 (−0.60)	0.15 (0.94)	0.07 (0.63)	−0.07 (−0.54)						0.86*** (6.29)

Table 6: Social transmission and the BTC sensitivity premium

This table reports analysis on the relation between social transmission and the *BTC sensitivity* premium. We reports excess returns (in percentage) of monthly rebalanced decile portfolios sorted on *BTC sensitivity* for stocks with and without concurrent tweets with “bitcoin” or “BTC” during the three-month formation period. “H – L” is the difference between the high and low portfolio’s excess return. Alpha is the four-factor alpha of a zero-cost portfolio that buys stocks in the top *BTC sensitivity* decile portfolio and sells stocks in the bottom *BTC sensitivity* decile portfolio. Column (1) reports the results for stocks without concurrent bitcoin tweets. Column (2) reports the results for stocks with concurrent bitcoin tweets. Column (3) reports results using the number of tweets for each stock as the portfolio weight to calculate the portfolio returns. The sample period is from January 2014 to September 2019. The t-statistics (in parentheses) are computed using Newey and West (1987) standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests.

	(1)	(2)	(3)
	Stocks w/o bitcoin tweets	Stocks w/ bitcoin tweets	Tweet-weighted returns
1 (Low)	0.63	0.40	−0.25
2	0.52	1.16	1.04
3	0.52	0.71	0.75
4	0.71	−0.24	0.38
5	0.52	0.50	−0.08
6	0.57	0.65	1.11
7	0.58	0.17	0.39
8	0.21	0.34	−1.36
9	0.16	−0.17	3.85
10 (High)	−0.71	−1.92	−4.01
H – L	−1.34***	−2.32**	−3.75***
t-stat	(−3.37)	(−2.00)	(−4.29)
Alpha	−1.31***	−2.60**	−4.01***
t-stat	(−3.83)	(−2.18)	(−4.78)

Table 7: Additional tests of social transmission

This table reports additional analysis on the social transmission and the *BTC sensitivity* premium. Panel A reports the monthly rebalanced long-short portfolio returns (in percentage) following high- and low-Google search periods for *BTC sensitivity* and three lottery-type stock characteristics: *MAX*, *IVOL*, and *EIS*. Panel B reports the monthly rebalanced long-short portfolio returns (in percentage) following high- and low stock market sentiment periods for *BTC sensitivity* and three lottery-type stock characteristics: *MAX*, *IVOL*, and *EIS*. Panel C reports the monthly rebalanced long-short portfolio returns (in percentage) following high- and low stock market sentiment periods for *BTC sensitivity*. We separate the sample into stocks with high and low retail trading activity. The sample period is from January 2014 to September 2019. The t-statistics (in parentheses) are computed using Newey and West (1987) standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests.

Panel A: Long-short BTC sensitivity portfolio return following high/low Google search periods.				
	BTC Sensitivity	MAX	IVOL	EIS
H–L following high Google search	–1.72*** (–4.71)	–1.95*** (–4.18)	–2.21*** (–5.79)	–2.78*** (–5.38)
H–L following low Google search	–0.89*** (–2.81)	–1.12*** (–2.56)	–1.29*** (–2.59)	–1.07** (–2.25)
Difference (t-stat)	–0.83** (–2.17)	–0.83 (–1.63)	–0.92* (–1.81)	–1.70*** (–4.74)
Panel B: Long-short BTC sensitivity portfolio return following high/low market sentiment periods				
	BTC Sensitivity	MAX	IVOL	EIS
H–L following high market sentiment	–1.17** (–2.49)	–1.59*** (–2.60)	–1.87*** (–2.81)	–2.12*** (–3.07)
H–L following low market sentiment	–1.24*** (–5.33)	–1.19*** (–3.49)	–1.35*** (–3.71)	–1.38*** (–2.79)
Difference (t-stat)	0.07 (0.25)	–0.41 (–0.97)	–0.52 (–1.02)	–0.74* (–1.78)
Panel C: Retail trading activity				
	High retail trading activity		Low retail trading activity	
H–L following high Google search	–1.90*** (–6.42)		–0.47 (–1.46)	
H–L following low Google search	–0.84** (–2.04)		–0.12 (–0.64)	
Difference (t-stat)	–1.06*** (–3.69)		–0.35*** (–3.18)	

Table 8: Instrumental variables analysis: First-stage regression of bitcoin return and Tether flow

This table reports the first-stage estimate of the relation between bitcoin return and Tether authorization and Tether flow. The dependent variable is the average three-hour bitcoin returns. $Flow(h-1)$ is the lagged average net hourly flow of Tether from Bitfinex to Poloniex and Bittrex and of bitcoin from Poloniex and Bittrex to Bitfinex. $AuthPost(h-1)$ is an indicator variable that equals one if a Tether authorization occurred in the previous 72 hours, and zero otherwise. $BTC(h-1)$ is the lagged realized bitcoin return. $VOB(h-1)$ is volatility calculated using hourly bitcoin returns over the previous 24 hours. All the above variables are at hourly frequency. We also control for the daily level Fama-French three factors: $MKT(t-1)$, $SMB(t-1)$, and $HML(t-1)$, lagged by one day. We suppress the intercept from the regression to conserve space. The t-statistics (in parentheses) are computed based on Newey and West (1987) standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from March 01, 2017 to March 31, 2018, which represents 396 days or 9504 hours.

Dependent variable	$\frac{1}{3} \sum_0^2 BTC_{h+i}$
Flow($h-1$)	-0.37 (-0.57)
AuthPost($h-1$)	-6.23*** (-3.07)
Flow($h-1$) $\times$ AuthPost($h-1$)	4.25** (2.53)
BTC($h-1$)	-0.01 (-0.48)
VOB($h-1$)	102.57*** (2.74)
BTC($h-1$) $\times$ VOB($h-1$)	-0.32 (-1.33)
MKT($t-1$)	-52.70 (-0.41)
SMB($t-1$)	-80.74 (-0.51)
HML($t-1$)	-498.79*** (-3.12)
Nobs	9,504
Adjusted R ²	0.01
F-stat	9.28

INTERNET APPENDIX
for
Does Cryptocurrency Trading Have Spillover Effects on
the Stock Market?

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This is the Internet Appendix for Does Cryptocurrency Trading Have Spillover Effects on the Stock Market? Section 1 describes our data sources and construction of the cryptocurrency market return. Section 2 reports additional results discussed in the main text. Section 3 examines whether the crypto-trading reduces the lottery premium.

1. Data description of bitcoin data

Figure A1 shows a summary of the entire cryptocurrency market from a screenshot of the CoinMarketCap.com home page on July 6, 2018. There are 1,615 cryptocurrencies with an estimated market capitalization of US\$266 billion. Table A1 lists the detailed top-ten coins used for the construction of the entire cryptocurrency market. The market capitalization is based on information on March 31, 2018.

Bitcoin is the largest cryptocurrency currently, followed by Ethereum, Ripple, Bitcoin Cash, and Litecoin. As reported in CoinMarketCap.com, bitcoin accounts for 43.92% market share of the total cryptocurrency market as of March 31, 2018, while the total market share of the second to tenth coin is 35.98%, which is less than the market share of bitcoin. Overall, the top-ten coins capture nearly 80% of the cryptocurrency market. Bitcoin and Litecoin have the oldest data availability; both are active since the first record date in CoinMarketCap.com, followed by Ripple and Stellar. Other cryptocurrencies, such as IOTA, EOS, Bitcoin Cash, and Cardona, have a relative short period of price data.

For the top-ten coins based on the market capitalization on March 31th, 2018, we extract data on history prices, available supply volume, trading volume over last 24 hours, and market capitalization. After calculating the corresponding stock-trading day's returns for all ten coins, we calculate the market capitalization weighted daily return as the market return of the entire cryptocurrency market.

2. Tables for additional test results

Table IA.1. Top 10 Cryptocurrencies

Table IA.2. Portfolio sorts: Alternative samples

Table IA.3. Portfolio sorts: Using value-weighted portfolios

Table IA.4. Additional Fama-MacBeth analysis

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Table IA.7. Portfolio sorts: Using CoinmarketCap.com sample

Table IA.8. Portfolio sorts: Using prominent digital currency market daily returns

Table IA.9. Portfolio sorts: Using 1-month, 6-month, and PCA based BTC sensitivity

Table IA.10. Portfolio sorts: Using Dimson adjustment

Table IA.11. Portfolio sorts: Using previous month return as portfolio weighting scheme

Table IA.12. Portfolio sorts: Out-of-sample test using financial stocks

Table IA.13. Fama-MacBeth regression results from positive and negative BTC sensitivity

Table IA.14. Does the crypto-trading reduce the negative lottery premium?

3. Does the crypto-trading reduce the negative lottery premium?

In this section, we further examine whether the spillover effect could be from stocks to bitcoin. If so, we expect there exists a possible substitution effect of skewness demand towards bitcoin and away from lottery stocks, and the lottery premium will become weaker. To examine this possibility that the crypto-trading reduce the interest in lottery stocks, we design the following test. At the end of each month t , we rank all stocks based on their lottery characteristic (*IVOL*, *MAX*, and *EIS*) and calculate a composite lottery index for each stock as the average percentile rank of the individual lottery characteristics. We take the top 10% of stocks in the composite lottery index, calculate an equal-weighted portfolio return of these stocks in month t , and compute the Spearman's rank correlation between the daily returns of the portfolio and bitcoin. We partition the sample into positive correlation and negative correlation periods and in the subsequent month ($t+1$), compute the raw return and four-factor alpha (the Fama-French three-factor model augmented by the momentum factor) of the long-short portfolio that buys stocks in the highest lottery composite index decile and shorts stocks in the lowest lottery composite index decile.

We argue that periods of positive correlation capture investors' common interest in bitcoin and lottery stocks, while periods of negative correlation reflect a shift in interest away from lottery stocks to cryptocurrencies. The prediction is that the negative lottery stock premium is state contingent that the negative lottery premium is large after positive correlation states and small after negative correlation states. Table IA.14 presents the results. Consistent with our prediction, we find that following periods of positive correlation, the long-short portfolio produces significantly negative returns but following periods of negative correlation, the long-short portfolio experiences insignificant returns.

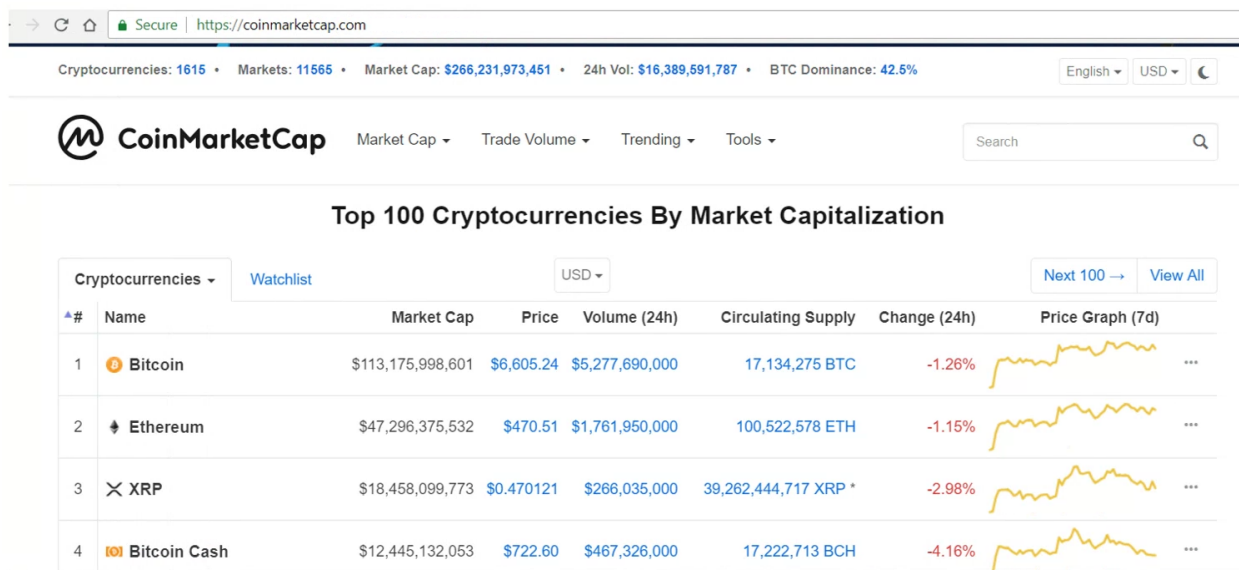


Figure IA.1: Screenshot of the homepage of CoinMarketCap.com

This screenshot is taken from coinmarketcap.com at about 23:00 pm on July 6, 2018. It shows the homepage of the website.



Figure IA.2: Sample tweets mentioning stocks and Bitcoin

In this figure, we show a screenshot of the sample tweets that mentions individual stocks and Bitcoin together.

Table IA.1: Top 10 Cryptocurrencies

This table shows the list of the 10 largest cryptocurrencies by market capitalization and their market share. Data are from CoinMarketCap.com as of March 31, 2018.

Rank	Coin name	Trading symbol	Data starting month	Market cap (\$B)	Market share
1	Bitcoin	BTC	201305	116.82	43.92%
2	Ethereum	ETH	201508	38.91	14.63%
3	Ripple	XRP	201308	19.96	7.50%
4	Bitcoin Cash	BCH	201708	11.89	4.47%
5	Litecoin	LTC	201305	6.63	2.49%
6	EOS	EOS	201707	4.65	1.75%
7	Cardona	ADA	201710	3.81	1.43%
8	Stellar	XLM	201408	3.51	1.32%
9	NEO	NEO	201609	3.27	1.23%
10	IOTA	MIOTA	201706	3.09	1.16%

Table IA.2: Portfolio sorts: Alternative samples

This table repeats the portfolio sorting analysis in Table 3 using alternative samples. The left panel includes our main sample and stocks with fundamental links with bitcoin/blockchain as identified by Autore et al. (2021). The right panel of all stocks further adds financial and utility stocks. At the end of each month, we sort the stocks into decile portfolio by their *BTC sensitivity* and estimate the portfolio returns. Excess return is the average returns minus the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include the CAPM, the Fama-French three-factor model, and the Fama-French three-factor augmented by the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that buy stocks in the top *BTC sensitivity* decile portfolio and sells stocks in the bottom *BTC sensitivity* decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	Including blockchain stocks				All stocks			
	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
Low	1.06 (2.83)	0.00 (−0.01)	0.16 (1.34)	0.21 (1.56)	1.12 (3.25)	0.12 (0.63)	0.28 (2.70)	0.31 (2.78)
2	1.02 (2.33)	−0.13 (−0.83)	0.02 (0.23)	0.06 (0.60)	1.05 (2.66)	−0.03 (−0.18)	0.12 (1.63)	0.16 (2.04)
3	0.95 (2.13)	−0.27 (−1.77)	−0.10 (−1.31)	−0.06 (−0.86)	0.94 (2.34)	−0.19 (−1.27)	−0.02 (−0.35)	0.01 (0.11)
4	0.97 (2.08)	−0.29 (−1.51)	−0.11 (−1.38)	−0.09 (−1.20)	1.09 (2.55)	−0.09 (−0.54)	0.07 (0.86)	0.10 (1.30)
5	0.99 (1.86)	−0.32 (−1.65)	−0.12 (−1.28)	−0.08 (−0.86)	1.01 (2.18)	−0.20 (−1.11)	−0.02 (−0.27)	0.01 (0.20)
6	0.98 (1.84)	−0.34 (−1.83)	−0.12 (−1.49)	−0.06 (−0.82)	0.98 (2.00)	−0.28 (−1.63)	−0.08 (−1.07)	−0.02 (−0.33)
7	0.87 (1.61)	−0.45 (−2.13)	−0.24 (−1.95)	−0.17 (−1.46)	0.99 (1.95)	−0.27 (−1.43)	−0.06 (−0.65)	−0.02 (−0.23)
8	0.77 (1.24)	−0.60 (−2.28)	−0.35 (−2.20)	−0.25 (−1.63)	0.75 (1.36)	−0.55 (−2.55)	−0.33 (−2.53)	−0.25 (−1.92)
9	0.47 (0.69)	−0.97 (−3.27)	−0.65 (−3.77)	−0.52 (−3.22)	0.58 (0.90)	−0.81 (−2.83)	−0.53 (−3.40)	−0.42 (−2.96)
High	−0.16 (−0.18)	−1.69 (−3.86)	−1.33 (−4.10)	−1.20 (−4.30)	−0.12 (−0.14)	−1.61 (−3.99)	−1.27 (−4.20)	−1.13 (−4.30)
H – L	−1.22**	−1.68***	−1.49***	−1.41***	−1.24**	−1.73***	−1.55***	−1.44***
(t-stat)	(−2.33)	(−3.96)	(−4.14)	(−4.29)	(−2.32)	(−4.02)	(−4.39)	(−4.58)

Table IA.3: Portfolio sorts: Value-weighted portfolios

This table repeats the portfolio sorting analysis in Table 3 using value-weighted portfolios. At the end of each month, we sort stock into decile portfolio by their *BTC sensitivity* and estimate the value-weighted portfolio returns. Excess return is the average returns minus the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include the CAPM, the Fama-French three-factor model, and the Fama-French three-factor augmented by the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that buy stocks in the top *BTC sensitivity* decile portfolio and sells stocks in the bottom *BTC sensitivity* decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	Value-weighted returns (Baseline sample)			
	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
Low	1.06 (3.51)	−0.07 (−0.57)	0.01 (0.05)	0.02 (0.22)
2	1.08 (2.98)	−0.05 (−0.47)	0.01 (0.12)	0.00 (−0.03)
3	1.04 (2.86)	−0.16 (−1.18)	−0.08 (−0.75)	−0.07 (−0.73)
4	1.07 (3.11)	−0.10 (−0.70)	0.01 (0.06)	0.02 (0.28)
5	1.09 (2.80)	−0.14 (−1.31)	−0.06 (−0.69)	−0.04 (−0.48)
6	1.06 (2.72)	−0.18 (−1.52)	−0.07 (−0.72)	−0.05 (−0.59)
7	0.83 (2.11)	−0.43 (−3.07)	−0.29 (−2.64)	−0.25 (−2.26)
8	0.77 (1.56)	−0.58 (−2.91)	−0.39 (−3.62)	−0.32 (−2.90)
9	0.67 (1.33)	−0.73 (−3.28)	−0.51 (−2.94)	−0.41 (−2.31)
High	0.09 (0.12)	−1.43 (−3.96)	−1.19 (−4.68)	−1.07 (−4.11)
H – L (t-stat))	−0.98** (−2.05)	−1.36*** (−3.45)	−1.19*** (−4.44)	−1.10*** (−4.16)

Table IA.4: Additional Fama MacBeth analysis

This table reports the time-series monthly averages of coefficient estimates from Fama-MacBeth cross-sectional regressions of returns. We standardize all stock characteristics ($N \sim (0,1)$) for ease of comparison. The dependent variable is a firm's monthly return (in percentage). The control variables include the logarithm firm of market capitalization ($Ln(SIZE)$), the logarithm of book-to-market equity ($Ln(B/M)$), annually operating profitability (OP), total asset growth (TAG), momentum (MOM), idiosyncratic volatility ($IVOL$), maximum daily return (MAX), expected idiosyncratic skewness (EIS), Amihud illiquidity ($ILLIQ$), average daily turnover ratio ($Turnover$), the logarithm of stock price ($Ln(Price)$), distress risk ($Distress$), short-term reversal (STR), institutional ownership ($IO\%$), retail trading activity (*Retail trading*), the number of analysts covering the stock (*No. analysts*), and analyst forecast dispersion (*Dispersion*). The t-statistics (in parentheses) are computed based on Newey and West (1987) standard errors with up to twelve lags. ***, **, and, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	(1)	(2)
BTC sensitivity	-0.14*** (-3.92)	-0.11*** (-3.32)
Ln(SIZE)	-0.07 (-0.80)	-0.36*** (-3.41)
Ln(B/M)	-0.03 (-0.21)	-0.04 (-0.31)
MOM	0.28*** (2.95)	0.32*** (3.36)
OP	0.33*** (4.27)	0.29*** (3.85)
TAG	-0.14* (-1.76)	-0.11 (-1.51)
IVOL	-0.47*** (-4.43)	-0.45*** (-4.68)
MAX	0.11 (1.08)	0.08 (0.86)
EIS	-0.47*** (-2.97)	-0.43** (-2.59)
ILLIQ	-0.03 (-0.96)	-0.03 (-1.00)
Turnover	-0.21** (-2.12)	-0.26** (-2.53)
Ln(Price)	-0.27 (-1.16)	-0.37* (-1.81)
Distress	0.24*** (3.33)	0.22*** (2.84)
STR		-0.02 (-0.28)
IO%		0.26*** (3.23)
Retail trading		-0.18** (-2.26)
No. analysts		0.21*** (3.09)
Dispersion		-0.10** (-2.06)

Table IA.5: Bitcoin factor risk-adjusted returns

This table presents alpha estimates and factor loadings of the long-short BTC sensitivity portfolio. In each month, all stocks are sorted by the absolute value of Bitcoin sensitivity into decile portfolios. We form a zero-cost portfolio that longs the bottom decile portfolio and shorts the top decile portfolio. Alpha is the intercept from the regression of the zero-cost portfolio on the respective model's risk factors. The factors include the four-factor model (the Fama-French three-factor model augmented by the momentum factor) plus one of the three factors extracted from the Bitcoin market: (i) the monthly Bitcoin excess return (BTC return), (ii) the monthly change in the hash rate (Δ BTC hash rate), and (iii) the monthly change in the number of Bitcoin transactions (Δ BTC transactions). The t-statistics (in parentheses) are computed based on Newey and West (1987) standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	(1)	(2)	(3)	(4)
Alpha	-1.50*** (-4.38)	-1.66*** (-4.10)	-1.45*** (-4.23)	-1.66*** (-4.04)
BTC return	0.00 (-0.88)			0.00** (-2.02)
Δ BTC hash rate		0.01 (0.72)		0.02* (1.77)
Δ BTC transaction			-0.02*** (-3.77)	-0.02*** (-3.88)
MKT	0.27*** (2.87)	0.26*** (2.85)	0.26*** (2.64)	0.26*** (2.64)
SMB	0.79*** (3.04)	0.78*** (3.02)	0.81*** (3.14)	0.80*** (3.07)
HML	-0.26*** (-2.82)	-0.26*** (-2.80)	-0.26*** (-2.89)	-0.27*** (-2.95)
UMD	-0.14 (-0.78)	-0.15 (-0.84)	-0.13 (-0.72)	-0.14 (-0.79)

Table IA.6: Portfolio sorts: Excluding high-tech firms

This table reports the average returns or alphas in percentage and the corresponding t-statistics of monthly rebalanced decile portfolios sorted by their BTC sensitivity. We exclude the high-tech firms from the sample. High-tech firms are defined as the industry 32, 34, 35 and 36 in the Fama and French 48 industry classification. Industry 32 is the telecommunication industry; the industry 34 is the business services industry; the industry 35 is the computer industry; and the industry 36 is the electronic equipment (chips) industry. Excess return is the average returns in excess of the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include: the CAPM, the Fama-French three-factor model (3-factor), and the Fama-French three-factor model plus the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that longs the top BTC sensitivity decile portfolio and shorts the bottom BTC sensitivity decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
1 (Low)	1.04 (2.75)	−0.03 (−0.12)	0.16 (1.25)	0.21 (1.48)
2	0.89 (1.95)	−0.24 (−1.18)	−0.07 (−0.46)	−0.05 (−0.34)
3	0.85 (1.85)	−0.39 (−2.00)	−0.19 (−2.29)	−0.13 (−1.61)
4	0.97 (1.88)	−0.31 (−1.42)	−0.10 (−0.89)	−0.06 (−0.55)
5	0.95 (1.77)	−0.37 (−1.71)	−0.15 (−1.60)	−0.10 (−1.00)
6	0.92 (1.70)	−0.43 (−2.49)	−0.20 (−1.66)	−0.11 (−1.04)
7	0.75 (1.31)	−0.59 (−2.39)	−0.35 (−2.57)	−0.26 (−2.02)
8	0.67 (1.03)	−0.70 (−2.27)	−0.43 (−2.07)	−0.32 (−1.54)
9	0.24 (0.36)	−1.20 (−3.95)	−0.85 (−4.83)	−0.73 (−4.72)
10 (High)	−0.22 (−0.26)	−1.75 (−3.95)	−1.40 (−4.24)	−1.25 (−4.12)
H – L (t-stat)	−1.26** (−2.40)	−1.73*** (−3.88)	−1.56*** (−4.36)	−1.46*** (−4.28)

Table IA.7: Portfolio sorts: Using CoinMarketCap.com sample

This table reports the average returns or alphas in percentage and the corresponding t-statistics of monthly rebalanced decile portfolios sorted by BTC sensitivity. Bitcoin return is calculated using the stock market trading day buy-and-hold returns to match the returns in CRSP. The calculation of the Bitcoin returns matches the identical trading dates used to compute the daily stock returns. Excess return is the average returns minus the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include: the CAPM, the Fama-French three-factor model (3-factor), and a four-factor model that includes Fama-French three factors and the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that longs the top BTC sensitivity decile portfolio and shorts the bottom BTC sensitivity decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
1 (Low)	0.93 (2.06)	0.00 (0.01)	0.20 (1.45)	0.26 (1.72)
2	0.67 (1.33)	-0.31 (-1.66)	-0.13 (-1.13)	-0.10 (-0.81)
3	0.74 (1.36)	-0.35 (-1.66)	-0.14 (-1.63)	-0.09 (-1.12)
4	0.80 (1.36)	-0.31 (-1.28)	-0.10 (-0.97)	-0.08 (-0.72)
5	0.74 (1.11)	-0.41 (-1.60)	-0.18 (-1.44)	-0.14 (-1.14)
6	0.81 (1.25)	-0.37 (-1.67)	-0.12 (-1.15)	-0.07 (-0.69)
7	0.56 (0.83)	-0.60 (-2.10)	-0.36 (-2.14)	-0.29 (-1.90)
8	0.48 (0.63)	-0.73 (-2.15)	-0.45 (-2.15)	-0.35 (-1.76)
9	0.20 (0.23)	-1.04 (-2.57)	-0.68 (-3.00)	-0.58 (-3.01)
10 (High)	-0.49 (-0.49)	-1.84 (-3.51)	-1.46 (-3.93)	-1.36 (-3.97)
H – L (t-stat)	-1.42** (-2.27)	-1.84*** (-3.50)	-1.65*** (-4.23)	-1.62*** (-4.19)

Table IA.8: Portfolio sorts: Using prominent digital currency market daily returns

This table reports the average returns or alphas in percentage and the corresponding t-statistics of monthly rebalanced decile portfolios sorted by their absolute return sensitivity to the whole digital currency market daily returns. The digital currency market return is calculated using the value-weighted daily returns of the top 10 digital currencies. We calculate the daily return for each coin using the stock market trading day buy-and-hold return. Excess return is the average returns minus the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include: the CAPM, the Fama-French three-factor model (3-factor factor), and the Fama-French three-factor model plus the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that buys stocks in the top BTC sensitivity decile portfolio and shorts stocks in the bottom BTC sensitivity decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from August 2013 to December 2022.

	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
1 (Low)	0.94 (1.94)	0.01 (0.04)	0.21 (1.42)	0.26 (1.56)
2	0.70 (1.40)	-0.27 (-1.40)	-0.10 (-0.84)	-0.06 (-0.48)
3	0.79 (1.48)	-0.30 (-1.47)	-0.10 (-1.63)	-0.06 (-1.04)
4	0.89 (1.47)	-0.24 (-1.04)	-0.03 (-0.25)	0.01 (0.08)
5	0.79 (1.30)	-0.33 (-1.47)	-0.11 (-0.88)	-0.08 (-0.69)
6	0.69 (1.06)	-0.47 (-2.00)	-0.22 (-1.70)	-0.14 (-1.24)
7	0.49 (0.72)	-0.69 (-2.66)	-0.44 (-2.59)	-0.36 (-2.30)
8	0.50 (0.66)	-0.71 (-2.21)	-0.43 (-2.62)	-0.36 (-2.26)
9	0.10 (0.11)	-1.16 (-2.79)	-0.82 (-3.49)	-0.71 (-3.28)
10 (High)	-0.47 (-0.45)	-1.78 (-3.21)	-1.39 (-3.52)	-1.31 (-3.72)
H – L (t-stat)	-1.41** (-2.22)	-1.79*** (-3.23)	-1.60*** (-3.63)	-1.57*** (-3.70)

Table IA.9: Portfolio sorts: Using 1-month, 6-month, and PCA based BTC sensitivity

This table reports the average returns or alphas in percentage and the corresponding t-statistics of monthly rebalanced decile portfolios sorted by their BTC sensitivity. Bitcoin return is calculated using the stock market trading day buy-and-hold returns to match the returns in CRSP. The calculation of the Bitcoin returns matches the identical trading dates used to compute the daily stock returns. Alternative BTC sensitivity measure is estimated either using previous 1 month, 6 months, or the PCA measure based on 1-month, 3-month, and 6-month BTC sensitivity. Excess return is the average returns minus the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include: the CAPM, the Fama-French three-factor model (3-factor factor), and the Fama-French three-factor model plus the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that longs the top BTC sensitivity decile portfolio and shorts the bottom BTC sensitivity decile portfolio. The t-statistics (in parentheses) are computed based on Newey and West (1987) standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. To get the 6-month BTC sensitivity will reduce the sample period, and the sample period in this table starts in October 2010 until December 2022.

		Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
1-m BTC Sensitivity	Low	0.95 (2.26)	−0.09 (−0.43)	0.10 (0.78)	0.15 (1.07)
	High	−0.16 (−0.19)	−1.54 (−3.26)	−1.12 (−3.19)	−1.00 (−3.38)
	H – L	−1.11**	−1.45***	−1.22***	−1.14***
	(t-stat)	(−2.18)	(−3.33)	(−3.31)	(−3.45)
6-m BTC Sensitivity	Low	1.09 (2.70)	0.05 (0.26)	0.26 (2.91)	0.30 (3.03)
	High	−0.13 (−0.14)	−1.57 (−3.29)	−1.17 (−3.22)	−1.03 (−3.38)
	H – L	−1.22**	−1.62***	−1.42***	−1.32***
	(t-stat)	(−2.26)	(−3.61)	(−3.68)	(−3.97)
PCA based BTC Sensitivity	Low	1.10 (3.50)	0.21 (1.04)	0.36 (2.56)	0.38 (2.56)
	High	−0.45 (−0.51)	−1.83 (−3.89)	−1.43 (−3.98)	−1.29 (−4.13)
	H – L	−1.55**	−2.04***	−1.78***	−1.67***
	(t-stat)	(−2.48)	(−3.85)	(−4.12)	(−4.22)

Table IA.10: Portfolio sorts: Using Dimson adjustment

This table reports the average returns or alphas and the corresponding t-statistics of monthly rebalanced on decile portfolios sorted by their BTC sensitivity. Bitcoin return is calculated using the stock market trading day buy-and-hold returns to match the returns in CRSP. The calculation of the Bitcoin returns matches the identical trading dates used to compute the daily stock returns. Following Dimson (1979), we also control for the lead-and-lag bitcoin return into our baseline regression and use the absolute value of the summation of the three coefficients of the lead-lag bitcoin returns as the BTC sensitivity measure. Excess return is the average returns minus the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include: the CAPM, the Fama-French three-factor model (3-factor factor), and the Fama-French three-factor model plus the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that longs the top sensitivity decile portfolio and shorts the bottom sensitivity decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
1 (Low)	1.16 (2.82)	0.14 (0.61)	0.32 (2.71)	0.37 (2.91)
2	0.97 (2.19)	−0.22 (−1.06)	−0.03 (−0.39)	0.02 (0.21)
3	0.89 (1.94)	−0.35 (−2.02)	−0.16 (−2.04)	−0.12 (−1.55)
4	0.90 (1.87)	−0.39 (−2.26)	−0.20 (−2.59)	−0.16 (−2.04)
5	0.92 (1.78)	−0.39 (−2.09)	−0.17 (−2.19)	−0.07 (−0.84)
6	0.90 (1.74)	−0.38 (−1.87)	−0.16 (−1.30)	−0.09 (−0.75)
7	0.84 (1.48)	−0.51 (−2.12)	−0.27 (−2.10)	−0.21 (−1.74)
8	0.71 (1.13)	−0.68 (−2.66)	−0.41 (−2.78)	−0.30 (−2.19)
9	0.19 (0.27)	−1.29 (−4.36)	−1.01 (−5.46)	−0.93 (−5.48)
10 (High)	−0.10 (−0.11)	−1.56 (−3.57)	−1.21 (−4.06)	−1.06 (−3.85)
H – L (t-stat)	−1.25** (−2.44)	−1.70*** (−3.77)	−1.53*** (−4.66)	−1.43*** (−4.33)

Table IA.11: Portfolio sorts: Using previous month return as portfolio weighting scheme

This table reports the average returns or alphas and the corresponding t-statistics of monthly rebalanced on decile portfolios sorted by their BTC sensitivity. Bitcoin return is calculated using the stock market trading day buy-and-hold returns to match the returns in CRSP. The calculation of the Bitcoin returns matches the identical trading dates used to compute the daily stock returns. To address the noise concern in the portfolio return estimation, following Asparouhova, Bessembinder, and Kalcheva (2010), we use the stock's past one month raw return as the stock's weight in the portfolio. Excess return is the average returns minus the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include: the CAPM, the Fama-French three-factor model (3-factor factor), and the Fama-French three-factor model plus the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that longs the top sensitivity decile portfolio and shorts the bottom sensitivity decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
1 (Low)	1.03 (2.72)	−0.03 (−0.14)	0.15 (1.11)	0.20 (1.37)
2	0.92 (2.10)	−0.19 (−1.07)	−0.02 (−0.18)	0.00 (0.02)
3	0.93 (2.05)	−0.30 (−1.70)	−0.11 (−1.52)	−0.06 (−0.83)
4	0.95 (1.94)	−0.30 (−1.42)	−0.11 (−1.17)	−0.08 (−0.86)
5	0.93 (1.70)	−0.36 (−1.72)	−0.15 (−1.44)	−0.11 (−1.08)
6	0.94 (1.84)	−0.36 (−2.08)	−0.13 (−1.54)	−0.06 (−0.85)
7	0.75 (1.37)	−0.54 (−2.40)	−0.32 (−2.42)	−0.25 (−1.97)
8	0.66 (1.07)	−0.65 (−2.37)	−0.40 (−2.13)	−0.30 (−1.61)
9	0.40 (0.58)	−0.97 (−2.94)	−0.64 (−2.99)	−0.53 (−2.85)
10 (High)	−0.51 (−0.64)	−1.96 (−4.61)	−1.62 (−4.89)	−1.52 (−4.68)
H – L (t-stat)	−1.54*** (−3.18)	−1.93*** (−4.35)	−1.77*** (−4.72)	−1.72*** (−4.48)

Table IA.12: Portfolio sorts: Out-of-sample test using financial stocks

This table reports the average returns or alphas in percentage and the corresponding t-statistics of monthly rebalanced equal-weighted stock returns on decile portfolios sorted by their BTC sensitivity. The sample contains financial firms only. Excess return is the average returns in excess of the risk-free rate. Alpha is the intercept from the regression of monthly excess returns on risk factors specified by an asset pricing model. The factor models include: the CAPM, the Fama-French three-factor model (3-factor factor), and the Fama-French three-factor model plus the momentum factor (4-factor). H – L return or alpha is the return or alpha of a zero-cost portfolio that longs the top sensitivity decile portfolio and shorts the bottom sensitivity decile portfolio. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	Excess Return	CAPM Alpha	3-factor Alpha	4-factor Alpha
1 (Low)	1.27 (3.64)	0.47 (1.39)	0.64 (4.18)	0.65 (4.55)
2	0.92 (2.73)	0.09 (0.27)	0.27 (1.50)	0.29 (1.75)
3	1.18 (2.74)	0.22 (0.71)	0.43 (2.49)	0.45 (3.05)
4	0.97 (2.57)	0.02 (0.08)	0.20 (1.21)	0.24 (1.73)
5	1.20 (2.91)	0.21 (0.75)	0.40 (2.29)	0.44 (2.96)
6	1.14 (2.72)	0.16 (0.47)	0.34 (2.29)	0.40 (3.05)
7	1.27 (2.98)	0.25 (0.83)	0.43 (2.30)	0.44 (2.52)
8	0.86 (2.16)	−0.14 (−0.45)	0.04 (0.19)	0.06 (0.34)
9	0.95 (1.91)	−0.09 (−0.29)	0.08 (0.31)	0.10 (0.37)
10 (High)	0.13 (0.20)	−1.04 (−2.32)	−0.86 (−1.87)	−0.77 (−1.78)
H – L (t-stat)	−1.14** (−2.21)	−1.50*** (−2.65)	−1.50*** (−3.11)	−1.42*** (−3.07)

Table IA.13: Fama-MacBeth regression: Positive and negative BTC sensitivity

This table reports the results from Fama-MacBeth return regressions for the positive and negative *BTC sensitivity*. All stock characteristics are standardized to make the results comparable. The dependent variable is a firm's monthly return (in percentage). The explanatory variable of interest is either the positive *BTC sensitivity* or negative *BTC sensitivity*. The control variables include: idiosyncratic volatility (*IVOL*), the logarithm firm of market cap ($\ln(\text{SIZE})$), the logarithm of book-to-market equity ($\ln(B/M)$), annually operating profitability (*OP*), total asset growth (*TAG*), momentum (*MOM*), maximum daily return (*MAX*), expected idiosyncratic skewness (*EIS*), Short-term reversal (*STR*), Amihud illiquidity (*ILLIQ*), average daily turnover (*Turnover*), the logarithm of stock price ($\ln(\text{Price})$), and the distress risk (*Distress*). The table reports the time-series monthly averages of the estimated coefficients. The t-statistics (in parentheses) are computed using Newey and West (1987) corrected standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	(1)	(2)	(3)
Positive BTC sensitivity	-0.22*** (-3.74)		-0.22*** (-3.89)
Negative BTC sensitivity		0.02 (0.30)	-0.01 (-0.30)
$\ln(\text{SIZE})$	-0.09 (-0.94)	-0.08 (-0.85)	-0.09 (-0.97)
$\ln(B/M)$	-0.02 (-0.21)	-0.02 (-0.16)	-0.02 (-0.20)
OP	0.26*** (2.85)	0.26*** (2.83)	0.26*** (2.84)
TAG	0.32*** (4.01)	0.33*** (4.08)	0.32*** (4.02)
MOM	-0.14* (-1.79)	-0.14* (-1.78)	-0.14* (-1.81)
IVOL	-0.50*** (-4.98)	-0.55*** (-5.46)	-0.51*** (-4.99)
MAX	0.11 (1.22)	0.12 (1.28)	0.11 (1.25)
EIS	-0.47*** (-2.89)	-0.47*** (-2.91)	-0.47*** (-2.90)
STR	-0.07 (-0.97)	-0.07 (-0.93)	-0.07 (-0.93)
ILLIQ	-0.03 (-0.77)	-0.02 (-0.64)	-0.02 (-0.71)
Turnover	-0.22** (-2.18)	-0.23** (-2.38)	-0.22** (-2.19)
$\ln(\text{Price})$	-0.24 (-1.05)	-0.23 (-1.02)	-0.24 (-1.06)
Distress	0.22*** (2.79)	0.22*** (2.77)	0.22*** (2.79)

Table IA.14: Does cryptocurrency trading reduce the negative lottery premium?

This table examines a possible substitution effect of the spillover effect towards bitcoin and away from lottery stocks. At the end of each month t , we rank all the stocks based on their lottery characteristic ($IVOL$, MAX , and EIS) and calculate a composite lottery index for each stock as the average percentile rank of the individual lottery characteristics. We define the top 10% stocks in the composite lottery index as lottery stocks. We extract the historical daily lottery stock returns in month t , calculate equal-weighted portfolio return of these stocks, and compute the Spearman's rank correlation between the daily return of the lottery stock portfolio and bitcoin (i.e., correlation (t)). We partition the entire sample period into positive correlation periods and negative correlation periods and compute the (H-L) raw return and 4-factor alpha (the Fama-French three-factor model plus the momentum factor) of the long-short portfolio that buys stocks in the highest composite lottery index decile and shorts stocks in the lowest composite lottery index decile in the subsequent month ($t+1$). The t-statistics (in parentheses) are computed based on Newey and West (1987) standard errors with up to twelve lags. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively, using two-tailed tests. The sample period is from October 2010 to December 2022.

	Composite lottery index	
	H-L raw return ($t+1$)	H-L 4-factor alpha ($t+1$)
Correlation (t) > 0	-2.75*** (-6.37)	-3.28*** (-7.30)
Correlation (t) ≤ 0	-1.31 (-1.03)	-1.71* (-1.95)
Difference	1.44* (1.76)	1.56*** (2.60)